RAJALAKSHMI ENGINEERING COLLEGE

Affiliated to ANNA UNIVERSITY, Chennai

MOBILE APPLICATION DEVELOPMENT

# Laboratory Record Note Book

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SEMESTER ..... VI

ACADEMIC YEAR ..... 2025 -26

RAJALAKSHMI ENGINEERING COLLEGE

ENGINEERING COLLEGE

An AUTONOMOUS Institution

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# BONAFIDE CERTIFICATE

NAME .....PRIYADHARSHINI K.....

ACADEMIC YEAR ...2025-26..... SEMESTER .....VI.....BRANCH...B TECH IT...

UNIVERSITY REGISTER No. 2116231001156

Certified that this is the bonafide record of work done by the above student in the

MOBILE APPLICATION DEVELOPMENT CS23621 Laboratory during the year 2025 - 2026

Signature of Faculty - in - Charge

Submitted for the Practical Examination held on .....

**Internal Examiner** **External Examiner**

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# INTRODUCTION

**What is Kotlin?**

Kotlin is a modern, trending programming language that was released in 2016 by JetBrains. It has become very popular since it is compatible with Java (one of the most popular programming languages out there), which means that Java code (and libraries) can be used in Kotlin programs.

**Kotlin is used for:**

- Mobile applications (specially Android apps)
- Web development
- Server side applications
- Data science

**Why Use Kotlin?**

- Kotlin is fully compatible with Java
- Kotlin works on different platforms (Windows, Mac, Linux, Raspberry Pi, etc.)
- Kotlin is concise and safe
- Kotlin is easy to learn, especially if you already know Java
- Kotlin is free to use
- Big community/support

**Kotlin IDE**

The easiest way to get started with Kotlin, is to use an IDE.

An IDE (Integrated Development Environment) is used to edit and compile code.

In this chapter, we will use IntelliJ (developed by the same people that created Kotlin) which is free to download from https://www.jetbrains.com/idea/download/.

**Android's Kotlin-first approach**

At Google I/O 2019, we announced that Android development will be increasingly Kotlin-first, and we've stood by that commitment. Kotlin is an expressive and concise programming language that reduces common code errors and easily integrates into existing apps. If you're looking to build an Android app, we recommend starting with Kotlin to take advantage of its best-in-class features.
In an effort to support Android development using Kotlin, we co-founded the Kotlin Foundation and have ongoing investments in improving compiler performance and build speed. To learn more about Android's commitment to being Kotlin-first, see Android's commitment to Kotlin.

**Why is Android development Kotlin-first?**

We reviewed feedback that came directly from developers at conferences, our Customer Advisory Board (CAB), Google Developer Experts (GDE), and through our developer research. Many developers already enjoy using Kotlin, and the request for more Kotlin support was clear. Here's what developers appreciate about writing in Kotlin:

- **Expressive and concise:** You can do more with less. Express your ideas and reduce the amount of boilerplate code. 67% of professional developers who use Kotlin say Kotlin has increased their productivity.

- **Safer code:** Kotlin has many language features to help you avoid common programming mistakes such as null pointer exceptions. Android apps that contain Kotlin code are 20% less likely to crash.
- **Interoperable:** Call Java-based code from Kotlin, or call Kotlin from Java-based code. Kotlin is 100% interoperable with the Java programming language, so you can have as little or as much of Kotlin in your project as you want.
- **Structured Concurrency:** Kotlin coroutines make asynchronous code as easy to work with as blocking code. Coroutines dramatically simplify background task management for everything from network calls to accessing local data.

**What does Kotlin-first mean?**

When building new Android development tools and content, such as Jetpack libraries, samples, documentation, and training content, we will design them with Kotlin users in mind while continuing to provide support for using our APIs from the Java programming language.

**Components**

Components of an Android Application
Introduction to Activities in Android
Services in Android with Example
Content Providers in Android with Example
Broadcast Receiver in Android With Example

**TextView**

TextView in Kotlin
Dynamic TextView in Kotlin
AutoCompleteTextView in Kotlin
Dynamic AutoCompleteTextView in Kotlin
CheckedTextView in Kotlin
Dynamic CheckedTextView in Kotlin

**ScrollView**

ScrollView in Android
HorizontalScrollView in Kotlin
Dynamic ScrollView in Kotlin
DynamicHorizontal ScrollView in Kotlin

Other Views Dynamic ImageView in Kotlin

Android ListView in Kotlin
CardView in Android With Example
VideoView in Kotlin
Zigzag View in Android
ViewAnimator in Android with Example

**Button**
Button in Kotlin
RadioButton in Kotlin
ImageButton in Kotlin
ToggleButton in Kotlin
RadioGroup in Kotlin
CheckBox in Kotlin
Dynamic ImageButton in Kotlin
Dynamic RadioButton in Kotlin

**EditText**
Android EditText in Kotlin
Dynamic EditText in Kotlin

**Layouts**
Layouts in Android UI Design
Android UI Layouts
Android TableLayout in Kotlin
Android FrameLayout in Kotlin
Android RelativeLayout in Kotlin
Android ConstraintLayout in Kotlin
Android LinearLayout in Kotlin
Android Motion Layout in Kotlin

**Bar**
SeekBar in Kotlin
Dynamic SeekBar in Kotlin
Discrete SeekBar in Kotlin
RatingBar in Kotlin
Dynamic RatingBar in Kotlin
ProgressBar in Kotlin
Dynamic ProgressBar in Kotlin

**Switcher**
Switch in Kotlin

Dynamic Switch in Kotlin
TextSwitcher in Kotlin
Dynamic TextSwitcher in Kotlin
ImageSwitcher in Kotlin
Dynamic ImageSwitcher in Kotlin
Time and Date Picker
TimePicker in Kotlin

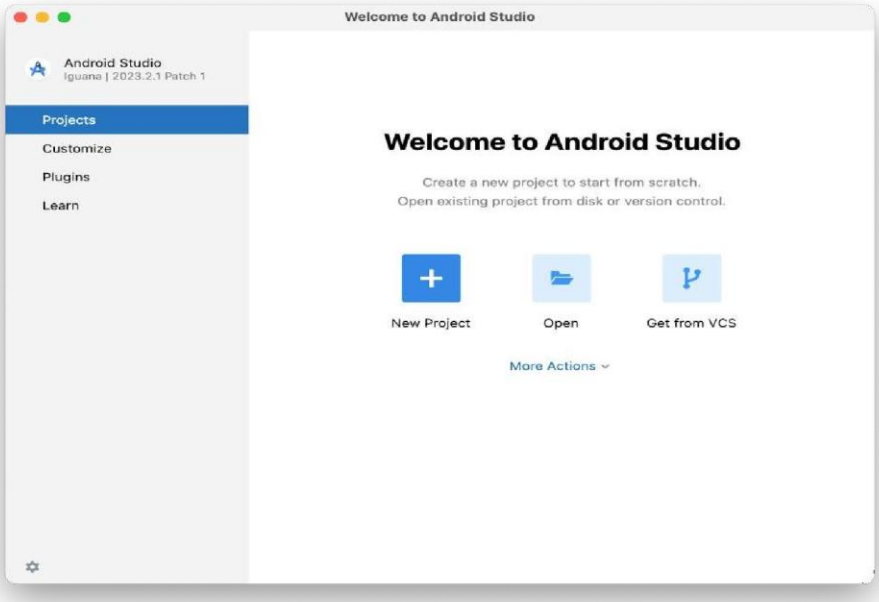
How to add a Custom-styled Toast in Android using Kotlin
Misc
Spinner
Dynamic Spinner in Kotlin
Advantage of Kotlin for Android Development
Choosing Kotlin for Android development offers several compelling benefits:

Concise Code: Kotlin lets you write less code to do the same things, making it easier to read and maintain.
Null Safety: Kotlin helps prevent common errors by making it harder to accidentally use null values, which can crash your app.
Interoperability with Java: You can use Kotlin and Java together in the same project.
Kotlin works well with existing Java code, so you don't have to rewrite everything.
Modern Features: Kotlin has many modern programming features that make development faster and more enjoyable, like extension functions and data classes.
Better Performance: Kotlin often results in better-performing apps due to its efficient syntax and features.
Official Support: Kotlin is officially supported by Google for Android development, ensuring good documentation and ongoing improvements.
Active Community: A growing community of developers means lots of resources, libraries, and tools to help you learn and solve problems.

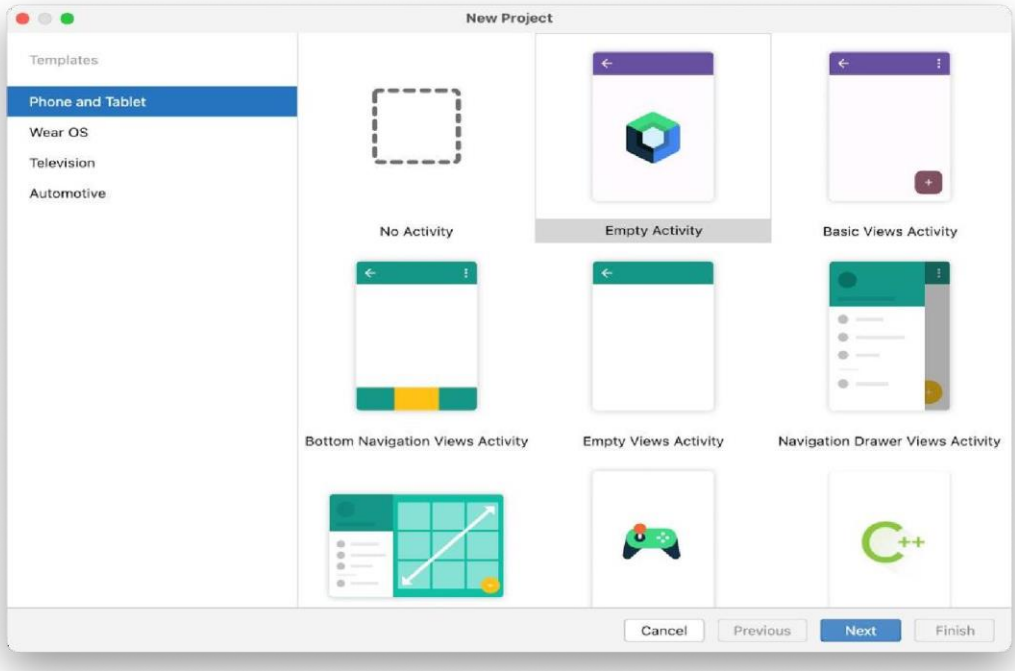
codelab, you create an Android app with the **Empty Activity** project template provided by Android Studio.
To create a project in Android Studio:

1. Double click the Android Studio icon to launch Android Studio.

2. In the **Welcome to Android Studio** dialog, click **New Project**.




The **New Project** window opens with a list of templates provided by Android Studio.

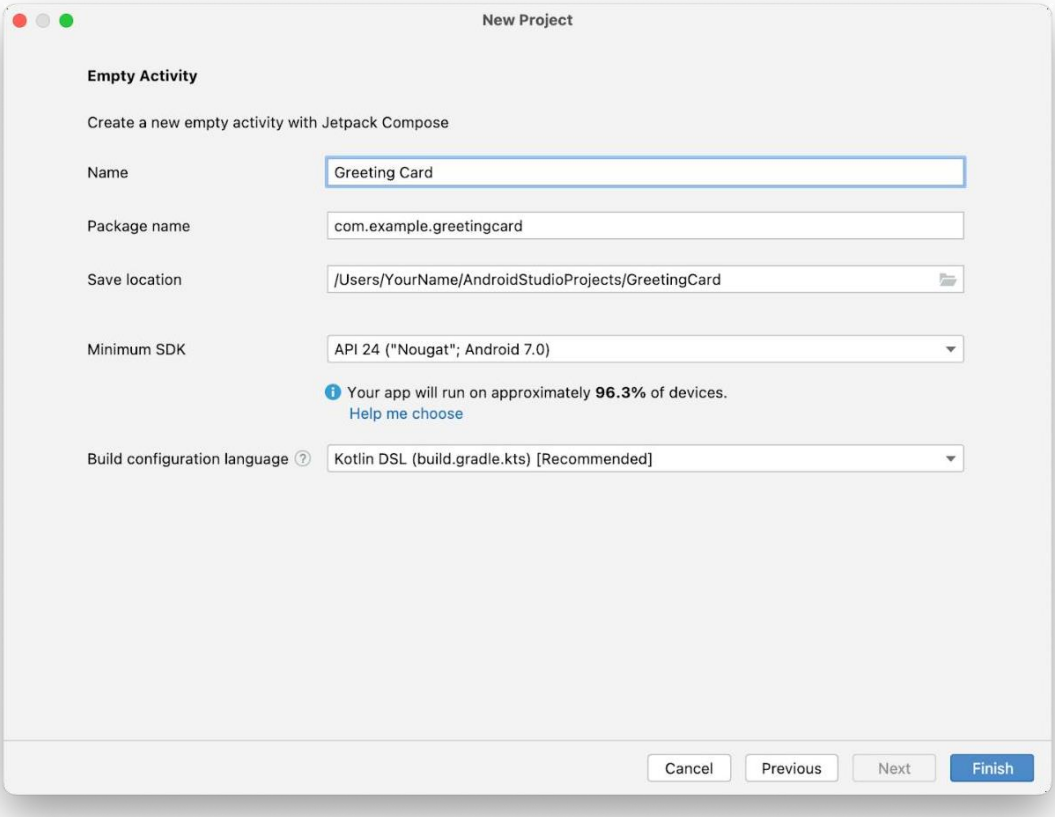



In Android Studio, a project template is an Android project that provides the blueprint for a certain type of app. Templates create the structure of the project and the files needed for Android Studio to build your project. The template that you choose provides starter code to get you going faster.

3. Make sure the **Phone and Tablet** tab is selected.
4. Click the **Empty Activity** template to select it as the template for your project.
   The **Empty Activity** template is the template to create a simple project that you can use to build a Compose app. It has a single screen and displays the text "Hello Android!".
5. Click **Next**. The **New Project** dialog opens. This has some fields to configure your project.
6. Configure your project as follows:

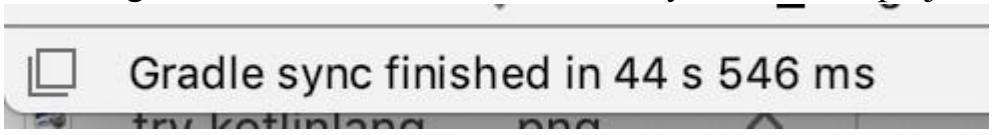
The **Name** field is used to enter the name of your project, for this codelab type "Greeting Card".

Leave the **Package name** field as is. This is how your files will be organized in the file structure. In this case, the package name will be com.example.greetingcard.
Leave the **Save location** field as is. It contains the location where all the files related to your project are saved. Take a note of where that is on your computer so that you can find your files.
Select **API 24: Android 7.0 (Nougat)** from the menu in the **Minimum SDK** field. Minimum SDK indicates the minimum version of Android that your app can run on.




7. Click **Finish**. This may take a while - this is a great time to get a cup of tea! While Android Studio is setting up, a progress bar and message indicates whether Android Studio is still setting up your project. It may look like this:

A message that looks similar to this informs you when the project set up is created.




8. You may see a **What's New** pane which contains updates on new features in Android Studio. Close it for now.

Assistant What's New

# What's New in Iguana

This panel describes some of the new features and behavior changes included i
update.

To open this panel again later, select **Help > What's New in Android Studio** from
main menu.

Read in a browser

## Version control system integration in App Quality Insights

Use **App Quality Insights** to navigate from a Crashlytics stack trace to the relev
code—at the point in time when the crash happened. When you view crash repo
choose to navigate to the line of code in your current git checkout or view a diff
between the current checkout and the version of your codebase that generated
crash.

Learn more

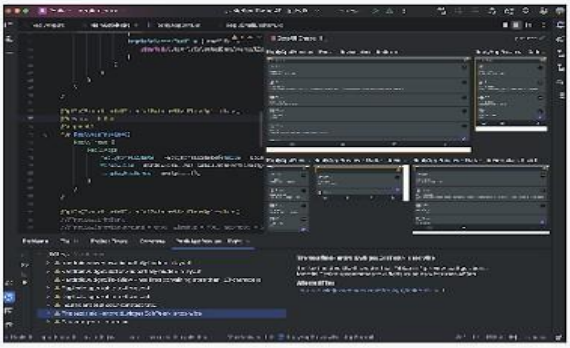
## View Crashlytics crash variants in App Quality Insights

To help you analyze the root causes of a crash, use **App Quality Insights** to view
by issue *variants*, or groups of events that share similar stack traces.

Learn more

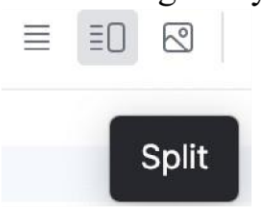
## UI Check Mode in Compose Preview

To help developers build more adaptive and accessible UI in Jetpack Compose,
Studio Iguana Canary 5 introduced a new UI Check mode in Compose Preview.
feature works similar to Visual Linting and Accessibility checks integrations for
Activate Compose UI check mode for Android Studio to automatically audit your
Compose UI and check for adaptive and accessibility issues across different sc
sizes, such as text stretched on large screens and low color contrast. The mode
highlights issues found in different Preview configurations and lists them in the
problems panel.

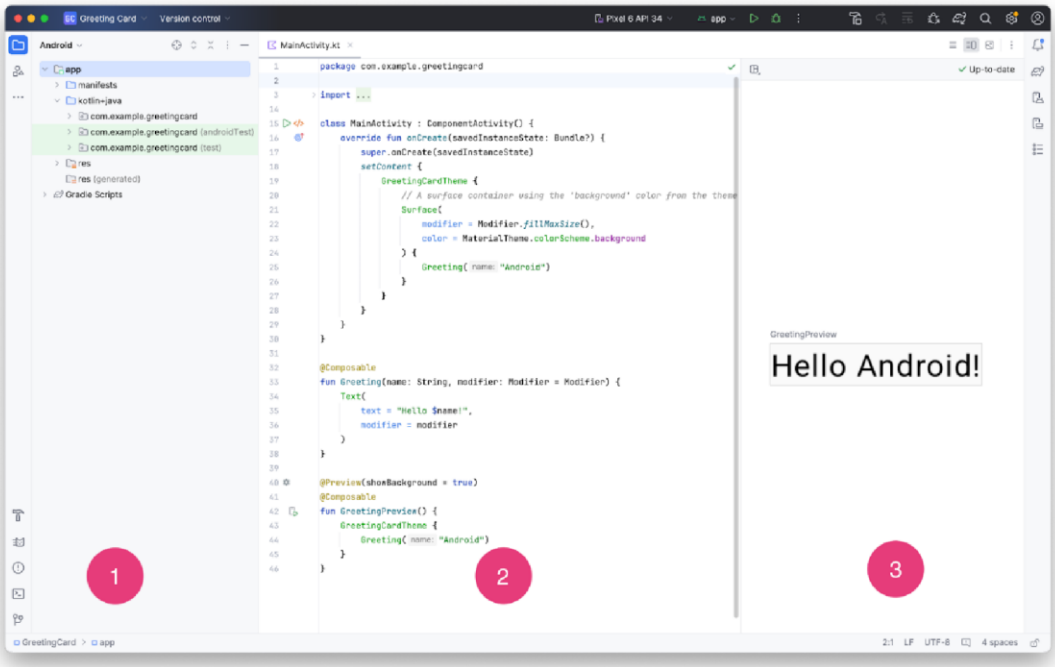


9. Click **Split** on the top right of Android Studio, this allows you to view both code and design. You can also click **Code** to view code only or click **Design** to view

design only.




After pressing **Split** you should see three areas:



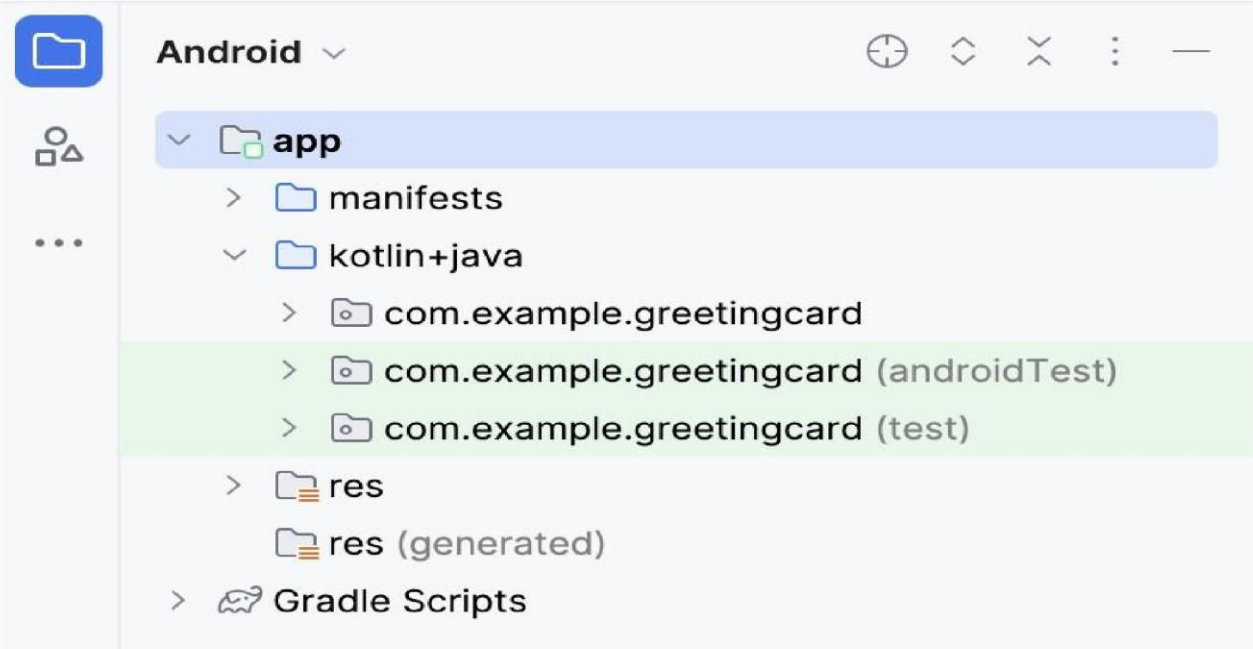

- The **Project** view (1) shows the files and folders of your project
- The **Code** view (2) is where you edit code
- The **Design** view (3) is where you preview what your app looks like

In the **Design** view, you may see a blank pane with this text:

10. Click **Build & Refresh**. It may take a while to build but when it is done the preview shows a text box that says "**Hello Android!**". Empty Compose activity contains all the code necessary to create this app.

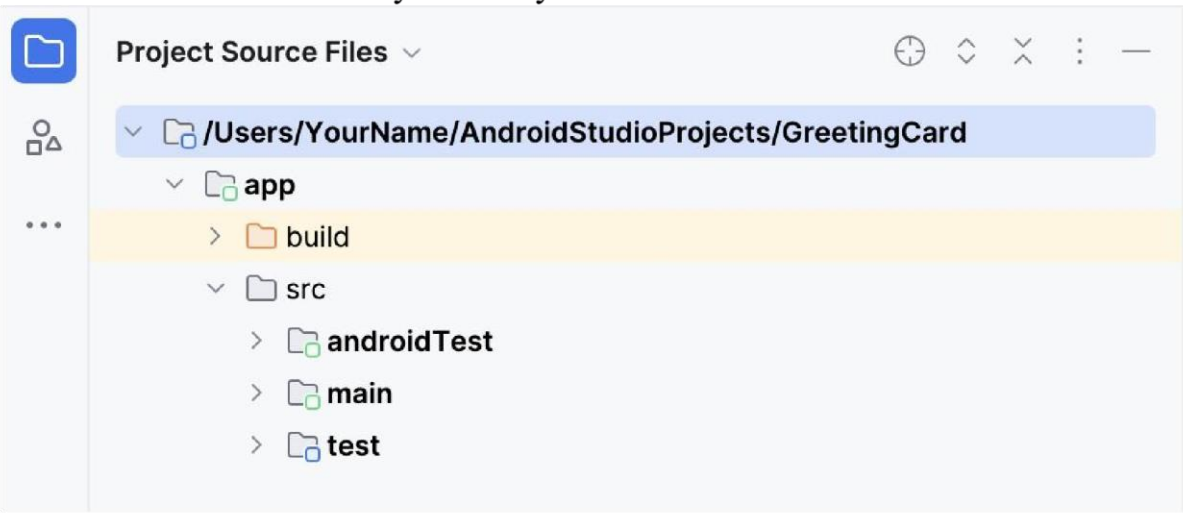
explore Android Studio by becoming familiar with the file structure.

1. In Android Studio, take a look at the **Project** tab. The **Project** tab shows the files and folders of your project. When you were setting up your project the package name was **com.example.greetingcard**. You can see that package right here in the **Project** tab. A package is basically a folder where code is located. Android Studio organizes the project in a directory structure made up of a set of packages.
2. If necessary, select **Android** from the drop-down menu in the **Project** tab.




This is the standard view and organization of files that you use. It's useful when you write code for your project because you can easily access the files you will be working on in your app. However, if you look at the files in a file browser, such as Finder or Windows Explorer, the file hierarchy is organized very differently.

3. Select **Project Source Files** from the drop-down menu. You can now browse the files in the same way as in any file browser.




4. Select **Android** again to switch back to the previous view. You use the **Android** view for this course. If your file structure ever looks strange, check to make sure you're still in **Android** view.

**Code** view of the MainActivity.kt file. Notice there are some automatically generated functions in this code, specifically the onCreate() and the setContent() functions.
**Note:** Remember that a function is a segment of a program that performs a specific task.

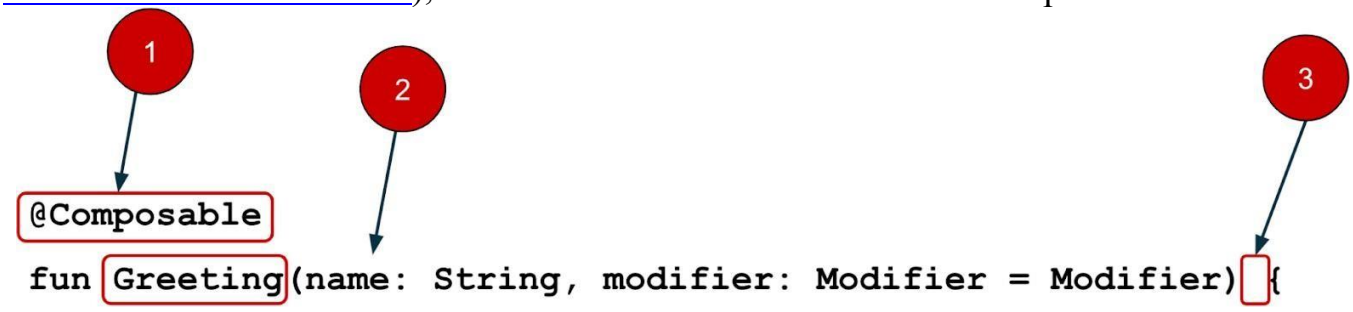
```
class MainActivity : ComponentActivity() {
```

```
        super.onCreate(savedInstanceState)
        setContent {
            GreetingCardTheme {
                // A surface container using the 'background' color from the theme
                Surface( modifier =
                    Modifier.fillMaxSize(),
                    color = MaterialTheme.colorScheme.background
                ) {
                    Greeting("Android")
                }
            }
        }
    }
}
```

The onCreate() function is the entry point to this Android app and calls other functions to build the user interface. In Kotlin programs, the main() function is the entry point/starting point of execution. In Android apps, the onCreate() function fills that role.
The setContent() function within the onCreate() function is used to define your layout through composable functions. All functions marked with the @Composable annotation can be called from the setContent() function or from other Composable functions. The annotation tells the Kotlin compiler that this function is used by Jetpack Compose to generate the UI.
**Note:** The compiler takes the Kotlin code you wrote, looks at it line by line, and translates it into something that the computer can understand. This process is called compiling your code.
Next, look at the Greeting() function. The Greeting() function is a Composable function, notice the @Composable annotation above it. This Composable function takes some input and generates what's shown on the screen.

```
@Composable
fun Greeting(name: String, modifier: Modifier = Modifier) {
    Text( text = "Hello
        $name!",
        modifier = modifier
    )
}
```

You've learned about functions before (if you need a refresher, refer to the Create and use functions in Kotlin codelab), but there are a few differences with composable functions.




.

1. You add the @Composable annotation before the function.

- 2. @Composable function names are capitalized.
- 3. @Composable functions can't return anything.

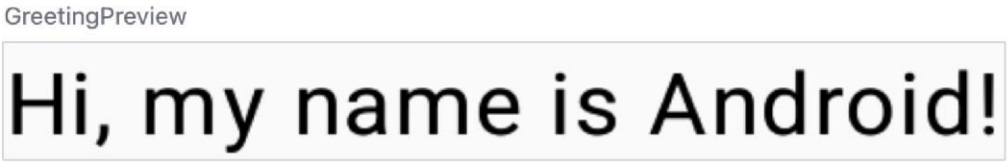
```
@Composable
fun Greeting(name: String, modifier: Modifier = Modifier) {
    Text(
        text = "Hello $name!",
        modifier = modifier
    )
}
```

Right now the Greeting() function takes in a name and displays Hello to that person.

1. Update the Greeting() function to introduce yourself instead of saying "Hello":

```
@Composable
fun Greeting(name: String, modifier: Modifier = Modifier) {
    Text(
        text = "Hi, my name is $name!",
        modifier = modifier
    )
}
```

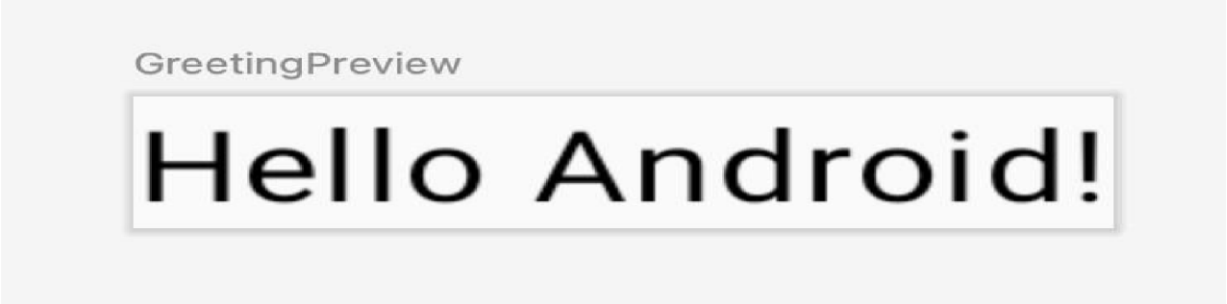
2. Android should automatically update the preview.




Great! You changed the text, but it introduces you as Android, which is probably not your name. Next, you will personalize it to introduce you with your name!

The GreetingPreview() function is a cool feature that lets you see what your composable looks like without having to build your entire app. To enable a preview of a composable, annotate with @Composable and @Preview. The @Preview annotation tells Android Studio that this composable should be shown in the design view of this file.

As you can see, the @Preview annotation takes in a parameter called showBackground. If showBackground is set to **true**, it will add a background to your composable preview. Since Android Studio by default uses a light theme for the editor, it can be hard to see the difference between showBackground = true and showBackground = false. However, this is an example of what the difference looks like. Notice the white background on the image set to true.



**Expt No : 1** **Date:**

# Kotlin Basics

## Aim

To study and implement basic Kotlin programming concepts such as variables, data types, operators, control statements, and null safety.

## Algorithm

1. Start the Android Studio IDE.
2. Create a new Kotlin project.
3. Declare variables using val and var.
4. Use basic data types such as Int, Double, String, and Boolean.
5. Implement conditional statements (if, when).
6. Implement loops (for, while).
7. Display output using println() or UI components.
8. Run the program and verify output.

**Coding** package com.example.basic1

```
import android.os.Bundle import androidx.activity.ComponentActivity import androidx.activity.compose.setContent import androidx.activity.enableEdgeToEdge import androidx.compose.foundation.layout.* import androidx.compose.foundation.rememberScrollState
import androidx.compose.foundation.verticalScroll import androidx.compose.material3.* import androidx.compose.runtime.* import androidx.compose.ui.Alignment import androidx.compose.ui.Modifier import androidx.compose.ui.text.font.FontWeight import androidx.compose.ui.unit.dp import androidx.compose.ui.unit.sp
import androidx.compose.ui.tooling.preview.Preview import com.example.basic1.ui.theme.Basic1Theme

// Function examples defined outside composable
fun add(a: Int, b: Int): Int = a + b fun greet(name: String, greeting: String = "Hello"): String = "$greeting, $name!" fun sumAll(vararg numbers: Int): Int = numbers.sum()
fun operate(a: Int, b: Int, operation: (Int, Int) -> Int): Int = operation(a, b)
```

```
tailrec fun factorial(n: Int, acc: Int = 1): Int = if (n <= 1) acc else factorial(n - 1, acc * n)

// Class examples defined outside composable
class Person(val name: String, var age: Int) { fun
introduce(): String = "Hi, I'm $name and I'm $age
years old"
} data class User(val id: Int, val username: String, val email:

String)

enum class Priority(val value: Int) {
   LOW(1), MEDIUM(2), HIGH(3), URGENT(4);
   fun getDisplayName(): String = name.lowercase().replaceFirstChar { it.uppercase() }
}

object AppConfig { const val APP_NAME
   = "Kotlin Demo"
   const val VERSION = "1.0.0"
   fun getConfig(): String = "$APP_NAME v$VERSION"
}

class MathUtils {
   companion object {
   const val PI = 3.14159
      fun circleArea(radius: Double): Double = PI * radius * radius
   }
}

abstract class Animal(val name: String) {
   abstract fun makeSound(): String fun
   sleep(): String = "$name is sleeping"
}

class Dog(name: String) : Animal(name) { override fun
makeSound(): String = "$name says: Woof!" }

interface Drawable { fun
   draw(): String
}

class Circle(private val radius: Double) : Drawable { override fun
   draw(): String = "Drawing a circle with radius $radius"
}
```

```
class MainActivity : ComponentActivity() { override
    fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContent {
            Basic1Theme {
                Scaffold(modifier = Modifier.fillMaxSize()) { innerPadding ->
                    KotlinFundamentalsDemo(
                        modifier = Modifier.padding(innerPadding)
                    )
                }
            }
        }
    }
}

@Composable
fun KotlinFundamentalsDemo(modifier: Modifier = Modifier) {
    var result by remember { mutableStateOf("") }

    Column( modifier =
        modifier
            .fillMaxSize()
            .verticalScroll(rememberScrollState())
            .padding(16.dp),        verticalArrangement        =
        Arrangement.spacedBy(16.dp)
    ) {
        Text( text = "Kotlin Fundamentals
            Demo", fontSize = 24.sp,
            fontWeight = FontWeight.Bold
        )

        // Variables and Data Types
        SectionCard(title = "Variables & Data Types") {
            demonstrateVariablesAndDataTypes()
        }

        // Operators
        SectionCard(title = "Operators") {
            demonstrateOperators()
        }

        // Control Statements
        SectionCard(title = "Control Statements") { demonstrateControlStatements()
```

```
        }

        // Null Safety
        SectionCard(title = "Null Safety") { demonstrateNullSafety()
        }

        // Functions
        SectionCard(title = "Functions") {
            demonstrateFunctions()
        }

        // Classes and Objects
        SectionCard(title = "Classes & Objects")
        { demonstrateClassesAndObjects() }

        // Interactive Example
        SectionCard(title = "Interactive Example") {
            InteractiveExample { newResult -> result
                = newResult
            } }

        if (result.isNotEmpty()) {
            Card( modifier =
                Modifier.fillMaxWidth(), colors =
                CardDefaults.cardColors(
                  containerColor = MaterialTheme.colorScheme.primaryContainer
                )
            ) {
                Text( text = "Result: $result", modifier
                    = Modifier.padding(16.dp), fontSize
                    = 16.sp
                )
            }
        }
    }
}

@Composable
fun SectionCard(title: String, content: @Composable () -> Unit) {
    Card( modifier =
        Modifier.fillMaxWidth(),
        elevation = CardDefaults.cardElevation(defaultElevation = 4.dp)
    ) {
```

```
        Column(                modifier                =
            Modifier.padding(16.dp),
            verticalArrangement = Arrangement.spacedBy(8.dp)
        ) {
            Text( text = title, fontSize = 18.sp,
                fontWeight = FontWeight.Bold,
                color = MaterialTheme.colorScheme.primary
            )
            content()
        }
    }
}

@Composable
fun demonstrateVariablesAndDataTypes() {
    //    Immutable    variable    (val)    val
    immutableName: String = "Kotlin"

    // Mutable variable (var) var
    mutableAge: Int = 10
    mutableAge += 1

    //      Type      inference      val
    inferredDouble = 3.14159

    // Different data types val byteValue: Byte = 127
    val shortValue: Short = 32767 val intValue: Int =
    2147483647      val      longValue:      Long      =
    9223372036854775807L
    val floatValue: Float = 3.14f val doubleValue:
    Double = 2.71828 val charValue: Char = 'K' val
    booleanValue: Boolean = true val stringValue:
    String = "Hello Kotlin" val arrayValue: IntArray =
    intArrayOf(1, 2, 3, 4, 5)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("val (immutable): $immutableName")
        Text("var (mutable): $mutableAge")
        Text("Type inference: $inferredDouble")
        Text("Byte: $byteValue")
        Text("Int: $intValue")
        Text("Long:  $longValue")  Text("Float:
        $floatValue")
        Text("Double: $doubleValue")
```

```
        Text("Char: $charValue")
        Text("Boolean: $booleanValue")
        Text("String: $stringValue")
        Text("Array: ${arrayValue.contentToString()}") }
}

@Composable
fun demonstrateOperators() {
    val a = 10 val
    b = 3 //
    Arithmetic
    operators val
    sum = a + b
    val
    difference =
    a - b val
    product = a *
    b val
    quotient = a /
    b
    val remainder = a % b

    // Comparison operators
    val isEqual = a == b val
    isNotEqual = a != b val
    isGreater = a > b
    val isLess = a < b

    // Logical operators val
    x = true val y = false val
    andResult = x && y val
    orResult = x || y val
    notResult = !x

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Arithmetic Operators (a=$a, b=$b):")
        Text(" a + b = $sum")
        Text(" a - b = $difference")
        Text(" a * b = $product")
        Text(" a / b = $quotient")
        Text(" a % b = $remainder")

        Text("Comparison Operators:")
        Text(" a == b = $isEqual")
```

```
        Text(" a != b = $isNotEqual")
        Text(" a > b = $isGreater")
        Text(" a < b = $isLess")

        Text("Logical Operators (x=$x, y=$y):")
        Text(" x && y = $andResult")
        Text(" x || y = $orResult")
        Text(" !x = $notResult")
    }
}

@Composable
fun demonstrateControlStatements() {
    val score = 85 val
    grade = when { score
    >= 90 -> "A" score
    >= 80 -> "B" score
    >= 70 -> "C" score
    >= 60 -> "D" else ->
    "F"
    }

    val numbers = listOf(1, 2, 3, 4, 5) val
    evenNumbers = mutableListOf<Int>()

    // For loop for (number in
    numbers) {
        if (number % 2 == 0) {
            evenNumbers.add(number)
        }
    }

    // While loop simulation var
    countdown = 5
    val countdownResult = buildString {
        while (countdown > 0) {
            append("$countdown ") countdown-
            -
        }
        append("Blast off!")
    }

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("If/When Statement:")
```

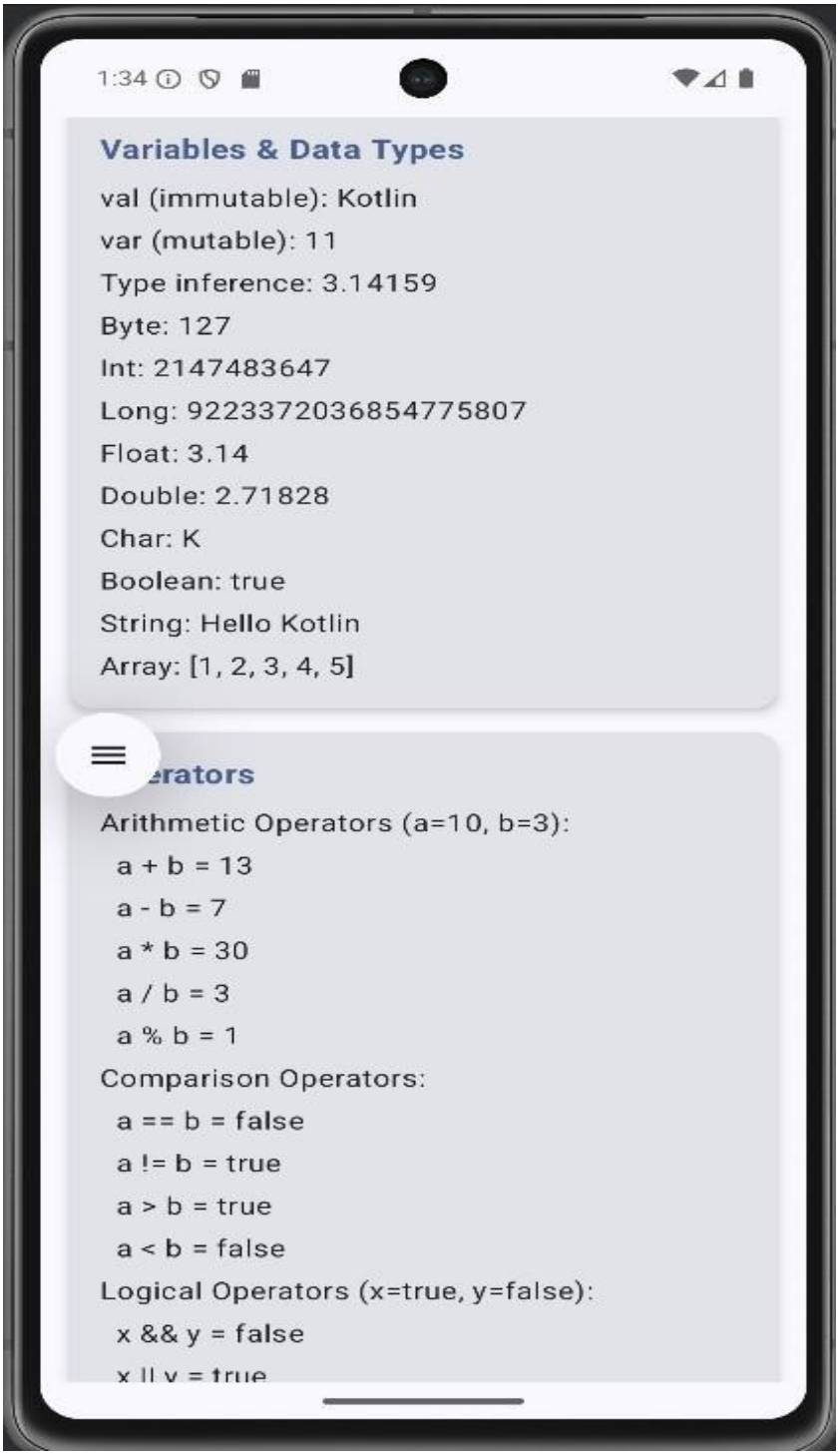
```
        Text(" Score: $score -> Grade: $grade")

        Text("For Loop:")
        Text(" Numbers: ${numbers.joinToString()}")
        Text(" Even numbers: ${evenNumbers.joinToString()}")

        Text("While Loop:")
        Text(" Countdown: $countdownResult")
    }
}

@Composable
fun demonstrateNullSafety() {
    // Nullable type
    val nullableString: String? = "Hello" val
    nullString: String? = null

    // Safe call operator val length1 =
    nullableString?.length

    // Elvis operator
    val length2 = nullString?.length ?: 0

    // Non-null asserted call (use with caution) val
    length3 = nullableString!!.length

    // Safe cast
    val anyValue: Any = "Kotlin" val
    safeCast = anyValue as? String

    // Let scope function val result =
    nullableString?.let {
        "Length of '$it' is ${it.length}"
    }

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Nullable String: $nullableString")
        Text("Null String: $nullString")
        Text("Safe call (?.): $length1")
        Text("Elvis (?:): $length2")
        Text("Non-null asserted (!!): $length3")
        Text("Safe cast (as?): $safeCast")
        Text("Let scope function: $result")
    }
```

```
}

@Composable
fun demonstrateFunctions() { //
   Lambda function
   val multiply = { a: Int, b: Int -> a * b }

   // Extension function fun String.isPalindrome():
   Boolean { val cleaned = this.lowercase().filter {
   it.isLetter() }
      return cleaned == cleaned.reversed()
   }

   val sum = add(5, 3) val greeting =
   greet("Kotlin") val sumVarargs = sumAll(1,
   2, 3, 4, 5) val product = operate(4, 5,
   multiply) val palindrome =
   "Racecar".isPalindrome() val fact =
   factorial(5)

   Column(verticalArrangement = Arrangement.spacedBy(4.dp)) { Text("Basic
      Function: add(5, 3) = $sum")
      Text("Default Parameter: greet(\"Kotlin\") = \"$greeting\"")
      Text("Varargs: sumAll(1,2,3,4,5) = $sumVarargs")
      Text("Higher-order: operate(4,5,multiply) = $product") Text("Extension:
      \"Racecar\".isPalindrome() = $palindrome")
      Text("Tail-recursive: factorial(5) = $fact") }
}

@Composable
fun demonstrateClassesAndObjects() {
   // Usage examples
   val person = Person("Alice", 25)
   val user = User(1, "john_doe", "john@example.com")
   val priority = Priority.HIGH val
   config = AppConfig.getConfig() val
   area = MathUtils.circleArea(5.0)
   val dog = Dog("Buddy") val
   circle = Circle(10.0)

   Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
      Text("Class: ${person.introduce()}")
      Text("Data class: User(username=${user.username}, email=${user.email})")
      Text("Enum: ${priority.getDisplayName()} = ${priority.value}")
```

```
        Text("Object: $config")
        Text("Companion object: Circle area (r=5) = $area")
        Text("Abstract class: ${dog.makeSound()}")
        Text("Interface: ${circle.draw()}") }
}

@Composable
fun InteractiveExample(onResult: (String) -> Unit) { var
    input1 by remember { mutableStateOf("") } var input2
    by remember { mutableStateOf("") } var operation by
    remember { mutableStateOf("add") }

    Column(verticalArrangement = Arrangement.spacedBy(8.dp)) { Text("Try
        it yourself!")

        OutlinedTextField( value
            = input1,
            onValueChange = { input1 = it }, label
            = { Text("Number 1") },
            modifier = Modifier.fillMaxWidth()
        )

        OutlinedTextField( value = input2,
            onValueChange = { input2 = it },
            label = { Text("Number 2") },
            modifier = Modifier.fillMaxWidth()
        )

        Row( horizontalArrangement =
            Arrangement.spacedBy(8.dp), modifier =
            Modifier.fillMaxWidth()
        ) {
            RadioButton( selected =
                operation == "add",
                onClick = { operation = "add" }
            )
            Text("Add")

            RadioButton( selected = operation ==
                "subtract",
                onClick = { operation = "subtract" }
            )
            Text("Subtract")
```

```
            RadioButton( selected = operation ==
                "multiply",
                onClick = { operation = "multiply" }
            )
            Text("Multiply")
        }

        Button( onClick
            = {
                val num1 = input1.toIntOrNull() ?: 0 val
                num2 = input2.toIntOrNull() ?: 0 val
                result = when (operation) {
                    "add" -> num1 + num2
                    "subtract" -> num1 - num2 "multiply"
                    -> num1 * num2
                    else -> 0
                }
                onResult("$num1 $operation $num2 = $result")
            },
            modifier = Modifier.align(Alignment.CenterHorizontally)
        ) {
            Text("Calculate")
        }
    }
}

@Preview(showBackground = true)
@Composable
fun KotlinFundamentalsDemoPreview() {
    Basic1Theme {
        KotlinFundamentalsDemo()
    }
}
```

**Output**




**Expt No : 2** **Date:**

## Functions

**Aim**

To understand and implement different types of functions in Kotlin.

**Algorithm**

1. Create a Kotlin file.
2. Define a function using fun keyword.

3. Pass parameters to the function.
4. Specify return type.
5. Call the function from main() or activity.
6. Implement default arguments.
7. Execute the program.

**Coding** package

com.example.basic1

import android.os.Bundle
import androidx.activity.ComponentActivity import
androidx.activity.compose.setContent import
androidx.activity.enableEdgeToEdge import
androidx.compose.foundation.layout.* import
androidx.compose.foundation.rememberScrollState
import androidx.compose.foundation.verticalScroll import
androidx.compose.material3.* import
androidx.compose.runtime.* import
androidx.compose.ui.Alignment import
androidx.compose.ui.Modifier import
androidx.compose.ui.text.font.FontWeight import
androidx.compose.ui.unit.dp import
androidx.compose.ui.unit.sp
import androidx.compose.ui.tooling.preview.Preview import
com.example.basic1.ui.theme.Basic1Theme

// Function examples defined outside composable
fun add(a: Int, b: Int): Int = a + b fun greet(name: String, greeting: String =
"Hello"): String = "$greeting, $name!" fun sumAll(vararg numbers: Int): Int =
numbers.*sum*()
fun operate(a: Int, b: Int, operation: (Int, Int) -> Int): Int = operation(a, b) tailrec fun
factorial(n: Int, acc: Int = 1): Int = if (n <= 1) acc else *factorial*(n - 1, acc * n)

// Class examples defined outside composable
class Person(val name: String, var age: Int) { fun
introduce(): String = "Hi, I'm $name and I'm $age
years old"
} data class User(val id: Int, val username: String, val email:

String)

enum class Priority(val value: Int) {
*LOW*(1), *MEDIUM*(2), *HIGH*(3), *URGENT*(4);
fun getDisplayName(): String = name.*lowercase*().*replaceFirstChar* { it.*uppercase*() }

```
}

object AppConfig { const val APP_NAME
   = "Kotlin Demo"
   const val VERSION = "1.0.0"
   fun getConfig(): String = "$APP_NAME v$VERSION"
}

class MathUtils {
   companion object {
   const val PI = 3.14159
      fun circleArea(radius: Double): Double = PI * radius * radius
   }
}

abstract class Animal(val name: String) {
   abstract fun makeSound(): String fun
   sleep(): String = "$name is sleeping"
}

class Dog(name: String) : Animal(name) { override fun
makeSound(): String = "$name says: Woof!" }

interface Drawable { fun
   draw(): String
}

class Circle(private val radius: Double) : Drawable { override fun
   draw(): String = "Drawing a circle with radius $radius"
}

class MainActivity : ComponentActivity() { override
   fun onCreate(savedInstanceState: Bundle?) {
   super.onCreate(savedInstanceState)
      enableEdgeToEdge()
      setContent {
         Basic1Theme {
            Scaffold(modifier = Modifier.fillMaxSize()) { innerPadding ->
               KotlinFundamentalsDemo(
                  modifier = Modifier.padding(innerPadding)
               )
            }
         }
      }
```

```
    }
}

@Composable
fun KotlinFundamentalsDemo(modifier: Modifier = Modifier) {
    var result by remember { mutableStateOf("") }

    Column( modifier =
        modifier
            .fillMaxSize()
            .verticalScroll(rememberScrollState())
            .padding(16.dp),        verticalArrangement        =
        Arrangement.spacedBy(16.dp)
    ) {
        Text( text = "Kotlin Fundamentals
            Demo", fontSize = 24.sp,
            fontWeight = FontWeight.Bold
        )

        // Variables and Data Types
        SectionCard(title = "Variables & Data Types") {
            demonstrateVariablesAndDataTypes()
        }

        // Operators
        SectionCard(title = "Operators") {
            demonstrateOperators()
        }

        // Control Statements
        SectionCard(title = "Control Statements") {
            demonstrateControlStatements()
        }

        // Null Safety
        SectionCard(title = "Null Safety") { demonstrateNullSafety()
        }

        // Functions
        SectionCard(title = "Functions") {
            demonstrateFunctions()
        }

        // Classes and Objects
```

```
        SectionCard(title = "Classes & Objects")
        { demonstrateClassesAndObjects() }

        // Interactive Example
        SectionCard(title = "Interactive Example") {
            InteractiveExample { newResult -> result
                = newResult
            } }

        if (result.isNotEmpty()) {
            Card(                modifier                =
                Modifier.fillMaxWidth(),   colors   =
                CardDefaults.cardColors(
                  containerColor = MaterialTheme.colorScheme.primaryContainer
                )
            ) {
                Text( text = "Result: $result", modifier
                    = Modifier.padding(16.dp), fontSize
                    = 16.sp
                )
            }
        }
    }
}

@Composable
fun SectionCard(title: String, content: @Composable () -> Unit) {
    Card(                modifier                =
        Modifier.fillMaxWidth(),
        elevation = CardDefaults.cardElevation(defaultElevation = 4.dp)
    ) {
        Column(                modifier                =
            Modifier.padding(16.dp),
            verticalArrangement = Arrangement.spacedBy(8.dp)
        ) {
            Text( text = title, fontSize = 18.sp,
                fontWeight = FontWeight.Bold,
                color = MaterialTheme.colorScheme.primary
            )
            content()
        }
    }
}
```

```
@Composable
fun demonstrateVariablesAndDataTypes() {
   // Immutable variable (val) val
   immutableName: String = "Kotlin"

   // Mutable variable (var) var
   mutableAge: Int = 10
   mutableAge += 1

   // Type inference val
   inferredDouble = 3.14159

   // Different data types val byteValue: Byte = 127
   val shortValue: Short = 32767 val intValue: Int =
   2147483647 val longValue: Long =
   9223372036854775807L
   val floatValue: Float = 3.14f val doubleValue:
   Double = 2.71828 val charValue: Char = 'K' val
   booleanValue: Boolean = true val stringValue:
   String = "Hello Kotlin" val arrayValue: IntArray =
   intArrayOf(1, 2, 3, 4, 5)

   Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
      Text("val (immutable): $immutableName")
      Text("var (mutable): $mutableAge")
      Text("Type inference: $inferredDouble")
      Text("Byte: $byteValue")
      Text("Int: $intValue")
      Text("Long: $longValue") Text("Float:
      $floatValue")
      Text("Double: $doubleValue")
      Text("Char: $charValue")
      Text("Boolean: $booleanValue")
      Text("String: $stringValue")
      Text("Array: ${arrayValue.contentToString()}") }
}

@Composable
fun demonstrateOperators() {
   val a = 10 val
   b = 3 //
   Arithmetic
   operators val
   sum = a + b
```

```
    val
    difference =
    a - b val
    product = a *
    b val
    quotient = a /
    b
    val remainder = a % b

    // Comparison operators
    val isEqual = a == b val
    isNotEqual = a != b val
    isGreater = a > b
    val isLess = a < b

    // Logical operators val
    x = true val y = false val
    andResult = x && y val
    orResult = x || y val
    notResult = !x

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Arithmetic Operators (a=$a, b=$b):")
        Text(" a + b = $sum")
        Text(" a - b = $difference")
        Text(" a * b = $product")
        Text(" a / b = $quotient")
        Text(" a % b = $remainder")

        Text("Comparison Operators:")
        Text(" a == b = $isEqual")
        Text(" a != b = $isNotEqual")
        Text(" a > b = $isGreater")
        Text(" a < b = $isLess")

        Text("Logical Operators (x=$x, y=$y):")
        Text(" x && y = $andResult")
        Text(" x || y = $orResult")
        Text(" !x = $notResult")
    }
}

@Composable
fun demonstrateControlStatements() {
```

```
    val score = 85 val
    grade = when { score
    >= 90 -> "A" score
    >= 80 -> "B" score
    >= 70 -> "C" score
    >= 60 -> "D" else ->
    "F"
    }

    val numbers = listOf(1, 2, 3, 4, 5) val
    evenNumbers = mutableListOf<Int>()

    // For loop for (number in
    numbers) {
        if (number % 2 == 0) {
            evenNumbers.add(number)
        }
    }

    // While loop simulation var
    countdown = 5
    val countdownResult = buildString {
        while (countdown > 0) {
            append("$countdown ") countdown-
            -
        }
        append("Blast off!")
    }

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("If/When Statement:")
        Text(" Score: $score -> Grade: $grade")

        Text("For Loop:")
        Text(" Numbers: ${numbers.joinToString()}")
        Text(" Even numbers: ${evenNumbers.joinToString()}")

        Text("While Loop:")
        Text(" Countdown: $countdownResult")
    }
}

@Composable
fun demonstrateNullSafety() {
```

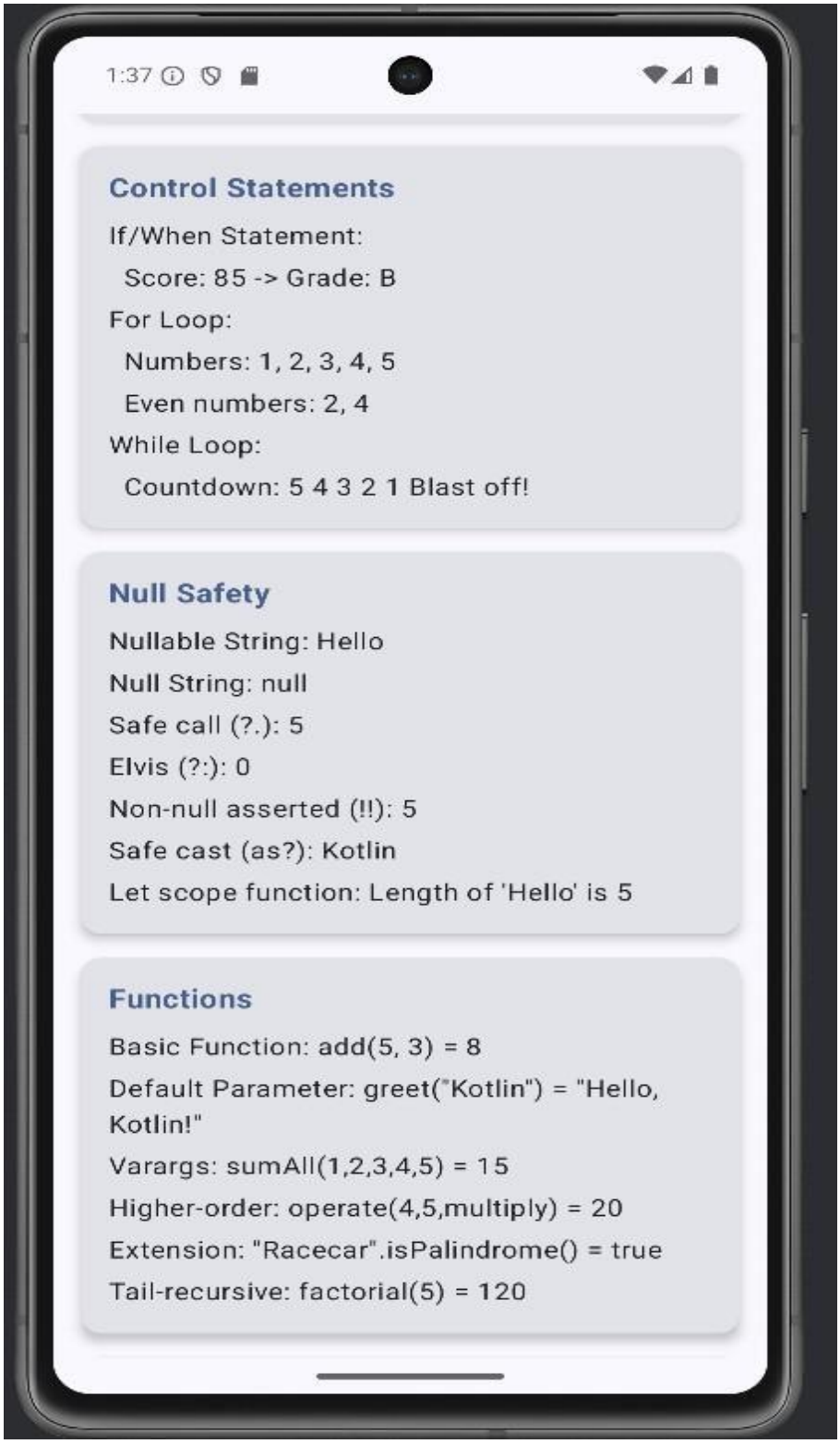
```
    // Nullable type
    val nullableString: String? = "Hello" val
    nullString: String? = null

    // Safe call operator val length1 =
    nullableString?.length

    // Elvis operator
    val length2 = nullString?.length ?: 0

    // Non-null asserted call (use with caution) val
    length3 = nullableString!!.length

    // Safe cast
    val anyValue: Any = "Kotlin" val
    safeCast = anyValue as? String

    // Let scope function val result =
    nullableString?.let {
        "Length of '$it' is ${it.length}"
    }

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Nullable String: $nullableString")
        Text("Null String: $nullString")
        Text("Safe call (?.): $length1")
        Text("Elvis (?:): $length2")
        Text("Non-null asserted (!!): $length3")
        Text("Safe cast (as?): $safeCast")
        Text("Let scope function: $result")
    }
}

@Composable
fun demonstrateFunctions() { //
    Lambda function
    val multiply = { a: Int, b: Int -> a * b }

    // Extension function fun String.isPalindrome():
    Boolean { val cleaned = this.lowercase().filter {
    it.isLetter() }
        return cleaned == cleaned.reversed()
    }
```

```
    val sum = add(5, 3) val greeting =
    greet("Kotlin") val sumVarargs = sumAll(1,
    2, 3, 4, 5) val product = operate(4, 5,
    multiply) val palindrome =
    "Racecar".isPalindrome() val fact =
    factorial(5)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) { Text("Basic
        Function: add(5, 3) = $sum")
        Text("Default Parameter: greet(\"Kotlin\") = \"$greeting\"")
        Text("Varargs: sumAll(1,2,3,4,5) = $sumVarargs")
        Text("Higher-order: operate(4,5,multiply) = $product") Text("Extension:
        \"Racecar\".isPalindrome() = $palindrome")
        Text("Tail-recursive: factorial(5) = $fact") }
}

@Composable
fun demonstrateClassesAndObjects() {
    // Usage examples
    val person = Person("Alice", 25)
    val user = User(1, "john_doe", "john@example.com")
    val priority = Priority.HIGH val
    config = AppConfig.getConfig() val
    area = MathUtils.circleArea(5.0)
    val dog = Dog("Buddy") val
    circle = Circle(10.0)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Class: ${person.introduce()}")
        Text("Data class: User(username=${user.username}, email=${user.email})")
        Text("Enum: ${priority.getDisplayName()} = ${priority.value}")
        Text("Object: $config")
        Text("Companion object: Circle area (r=5) = $area")
        Text("Abstract class: ${dog.makeSound()}")
        Text("Interface: ${circle.draw()}") }
}

@Composable
fun InteractiveExample(onResult: (String) -> Unit) { var
    input1 by remember { mutableStateOf("") } var input2
    by remember { mutableStateOf("") } var operation by
    remember { mutableStateOf("add") }
```

```
Column(verticalArrangement = Arrangement.spacedBy(8.dp)) { Text("Try
    it yourself!")

    OutlinedTextField( value
        = input1,
        onValueChange = { input1 = it }, label
        = { Text("Number 1") },
        modifier = Modifier.fillMaxWidth()
    )

    OutlinedTextField( value = input2,
        onValueChange = { input2 = it },
        label = { Text("Number 2") },
        modifier = Modifier.fillMaxWidth()
    )

    Row( horizontalArrangement =
        Arrangement.spacedBy(8.dp), modifier =
        Modifier.fillMaxWidth()
    ) {
        RadioButton( selected =
            operation == "add",
            onClick = { operation = "add" }
        )
        Text("Add")

        RadioButton( selected = operation ==
            "subtract",
            onClick = { operation = "subtract" }
        )
        Text("Subtract")

        RadioButton( selected = operation ==
            "multiply",
            onClick = { operation = "multiply" }
        )
        Text("Multiply")
    }

    Button( onClick
        = {
            val num1 = input1.toIntOrNull() ?: 0 val
            num2 = input2.toIntOrNull() ?: 0 val
            result = when (operation) {
```

```
                "add" -> num1 + num2
                "subtract" -> num1 - num2 "multiply"
                -> num1 * num2
                else -> 0
            }
            onResult("$num1 $operation $num2 = $result")
        },
        modifier = Modifier.align(Alignment.CenterHorizontally)
    ) {
        Text("Calculate")
    }
  }
}

@Preview(showBackground = true)
@Composable
fun KotlinFundamentalsDemoPreview() {
  Basic1Theme {
    KotlinFundamentalsDemo()
  }
}
```

**Output**




**Expt No : 3** **Date:**

## Classes and Objects

**Aim**

To implement object-oriented programming concepts using classes and objects in Kotlin.

**Algorithm**

1. Define a class using class keyword.
2. Declare properties inside the class.
3. Define member functions.
4. Create object using constructor.
5. Access class members.
6. Execute and verify output.

**Coding** package
com.example.basic1

import android.os.Bundle
import androidx.activity.ComponentActivity import
androidx.activity.compose.setContent import
androidx.activity.enableEdgeToEdge import
androidx.compose.foundation.layout.* import
androidx.compose.foundation.rememberScrollState
import androidx.compose.foundation.verticalScroll import
androidx.compose.material3.* import
androidx.compose.runtime.* import
androidx.compose.ui.Alignment import
androidx.compose.ui.Modifier import
androidx.compose.ui.text.font.FontWeight import
androidx.compose.ui.unit.dp import
androidx.compose.ui.unit.sp
import androidx.compose.ui.tooling.preview.Preview import
com.example.basic1.ui.theme.Basic1Theme

// Function examples defined outside composable
fun add(a: Int, b: Int): Int = a + b
fun greet(name: String, greeting: String = "Hello"): String = "$greeting, $name!" fun
sumAll(vararg numbers: Int): Int = numbers.*sum*()
fun operate(a: Int, b: Int, operation: (Int, Int) -> Int): Int = operation(a, b)
tailrec fun factorial(n: Int, acc: Int = 1): Int = if (n <= 1) acc else *factorial*(n - 1, acc * n)

// Class examples defined outside composable class Person(val
name: String, var age: Int) { fun introduce(): String = "Hi, I'm
$name and I'm $age years old"
}
data class User(val id: Int, val username: String, val email: String)

enum class Priority(val value: Int) {
  *LOW*(1), *MEDIUM*(2), *HIGH*(3), *URGENT*(4);
  fun getDisplayName(): String = name.*lowercase*().*replaceFirstChar* { it.*uppercase*() }
}

```
object AppConfig { const val APP_NAME
   = "Kotlin Demo"
   const val VERSION = "1.0.0"
   fun getConfig(): String = "$APP_NAME v$VERSION"
}

class        MathUtils        {
   companion     object     {
   const val PI = 3.14159
      fun circleArea(radius: Double): Double = PI * radius * radius
   }
}

abstract class Animal(val name: String) {
abstract fun makeSound(): String fun
sleep(): String = "$name is sleeping" }

class Dog(name: String) : Animal(name) { override fun
   makeSound(): String = "$name says: Woof!"
}

interface Drawable { fun
   draw(): String
}

class Circle(private val radius: Double) : Drawable { override fun
draw(): String = "Drawing a circle with radius $radius" }

class MainActivity : ComponentActivity() { override
   fun onCreate(savedInstanceState: Bundle?) {
   super.onCreate(savedInstanceState)
      enableEdgeToEdge()
      setContent {
         Basic1Theme {
            Scaffold(modifier = Modifier.fillMaxSize()) { innerPadding ->
               KotlinFundamentalsDemo(      modifier      =
                  Modifier.padding(innerPadding)
               )
            }
         }
      }
   }
}
```

```
@Composable
fun KotlinFundamentalsDemo(modifier: Modifier = Modifier) {
    var result by remember { mutableStateOf("") }

    Column( modifier =
        modifier
            .fillMaxSize()
            .verticalScroll(rememberScrollState())
            .padding(16.dp),        verticalArrangement        =
        Arrangement.spacedBy(16.dp)
    ) {
        Text( text = "Kotlin Fundamentals
            Demo", fontSize = 24.sp,
            fontWeight = FontWeight.Bold
        )

        // Variables and Data Types
        SectionCard(title = "Variables & Data Types") {
            demonstrateVariablesAndDataTypes()
        }

        // Operators
        SectionCard(title = "Operators") { demonstrateOperators()
        }

        // Control Statements
        SectionCard(title = "Control Statements") {
            demonstrateControlStatements()
        }

        // Null Safety
        SectionCard(title = "Null Safety") { demonstrateNullSafety()
        }

        // Functions
        SectionCard(title = "Functions") {
            demonstrateFunctions()
        }
        // Classes and Objects
        SectionCard(title = "Classes & Objects") { demonstrateClassesAndObjects()
        }

        // Interactive Example
        SectionCard(title = "Interactive Example") {
```

```
            InteractiveExample { newResult -> result
                = newResult
            }
        }

        if (result.isNotEmpty()) {
            Card(                modifier                =
                Modifier.fillMaxWidth(),   colors   =
                CardDefaults.cardColors(
                  containerColor = MaterialTheme.colorScheme.primaryContainer
                )
            ) {
                Text( text = "Result: $result", modifier
                    = Modifier.padding(16.dp), fontSize
                    = 16.sp
                )
            }
        }
    }
}

@Composable
fun SectionCard(title: String, content: @Composable () -> Unit) {
    Card(                modifier                =
        Modifier.fillMaxWidth(),
        elevation = CardDefaults.cardElevation(defaultElevation = 4.dp)
    ) {
        Column(                modifier                =
            Modifier.padding(16.dp),
            verticalArrangement = Arrangement.spacedBy(8.dp)
        ) {
            Text( text = title, fontSize = 18.sp,
                fontWeight = FontWeight.Bold,
                color = MaterialTheme.colorScheme.primary
            )
            content()
        }
    }
}

@Composable
fun demonstrateVariablesAndDataTypes() {
```

```
    // Immutable variable (val) val
    immutableName: String = "Kotlin"

    // Mutable variable (var)
    var mutableAge: Int = 10
    mutableAge += 1

    // Type inference
    val inferredDouble = 3.14159

    // Different data types val byteValue: Byte = 127
    val shortValue: Short = 32767 val intValue: Int =
    2147483647 val longValue: Long =
    9223372036854775807L
    val floatValue: Float = 3.14f val doubleValue:
    Double = 2.71828 val charValue: Char = 'K' val
    booleanValue: Boolean = true val stringValue:
    String = "Hello Kotlin" val arrayValue: IntArray =
    intArrayOf(1, 2, 3, 4, 5)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("val (immutable): $immutableName")
        Text("var (mutable): $mutableAge")
        Text("Type inference: $inferredDouble")
        Text("Byte: $byteValue")
        Text("Int: $intValue")
        Text("Long: $longValue") Text("Float:
        $floatValue")
        Text("Double: $doubleValue")
        Text("Char: $charValue")
        Text("Boolean: $booleanValue")
        Text("String: $stringValue")
        Text("Array: ${arrayValue.contentToString()}") }
}

@Composable
fun demonstrateOperators() {
    val a = 10 val
    b = 3

    // Arithmetic operators
    val sum = a + b val
    difference = a - b
```

```
    val product = a * b
    val quotient = a / b
    val remainder = a % b

    // Comparison operators
    val isEqual = a == b val
    isNotEqual = a != b val
    isGreater = a > b
    val isLess = a < b

    // Logical operators val
    x = true val y = false val
    andResult = x && y val
    orResult = x || y
    val notResult = !x

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Arithmetic Operators (a=$a, b=$b):")
        Text(" a + b = $sum")
        Text(" a - b = $difference")
        Text(" a * b = $product")
        Text(" a / b = $quotient")
        Text(" a % b = $remainder")

        Text("Comparison Operators:")
        Text(" a == b = $isEqual")
        Text(" a != b = $isNotEqual")
        Text(" a > b = $isGreater")
        Text(" a < b = $isLess")

        Text("Logical Operators (x=$x, y=$y):")
        Text(" x && y = $andResult")
        Text(" x || y = $orResult")
        Text(" !x = $notResult")
    }
}

@Composable
fun demonstrateControlStatements() {
    val  score  =  85  val
    grade = when { score
    >=  90  ->  "A"  score
    >=  80  ->  "B"  score
```

```
    >=  70  ->  "C"  score
    >= 60 -> "D" else ->
    "F"
    }

    val numbers = listOf(1, 2, 3, 4, 5)
    val evenNumbers = mutableListOf<Int>()

    // For loop for (number in
    numbers) {
        if (number % 2 == 0) {
            evenNumbers.add(number)
        }
    }

    // While loop simulation var
    countdown = 5
    val countdownResult = buildString {
        while (countdown > 0) {
            append("$countdown ") countdown-
            -
        }
        append("Blast off!")
    }

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("If/When Statement:")
        Text(" Score: $score -> Grade: $grade")

        Text("For Loop:")
        Text(" Numbers: ${numbers.joinToString()}")
        Text(" Even numbers: ${evenNumbers.joinToString()}")

        Text("While Loop:")
        Text(" Countdown: $countdownResult")
    }
}

@Composable
fun demonstrateNullSafety() {
    // Nullable type
    val nullableString: String? = "Hello" val
    nullString: String? = null
```

```
    // Safe call operator
    val length1 = nullableString?.length

    // Elvis operator
    val length2 = nullString?.length ?: 0
    // Non-null asserted call (use with caution) val
    length3 = nullableString!!.length

    // Safe cast
    val anyValue: Any = "Kotlin"
    val safeCast = anyValue as? String

    // Let scope function val result =
    nullableString?.let {
        "Length of '$it' is ${it.length}"
    }

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Nullable String: $nullableString")
        Text("Null String: $nullString")
        Text("Safe call (?.): $length1")
        Text("Elvis (?:): $length2")
        Text("Non-null asserted (!!): $length3")
        Text("Safe cast (as?): $safeCast")
        Text("Let scope function: $result")
    }
}

@Composable
fun demonstrateFunctions() { //
    Lambda function
    val multiply = { a: Int, b: Int -> a * b }

    // Extension function fun String.isPalindrome():
    Boolean { val cleaned = this.lowercase().filter {
    it.isLetter() }
        return cleaned == cleaned.reversed()
    }

    val sum = add(5, 3) val greeting =
    greet("Kotlin") val sumVarargs = sumAll(1,
    2, 3, 4, 5) val product = operate(4, 5,
    multiply) val palindrome =
```

```
    "Racecar".isPalindrome()     val     fact     =
    factorial(5)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) { Text("Basic
        Function: add(5, 3) = $sum")
        Text("Default Parameter: greet(\"Kotlin\") = \"$greeting\"")
        Text("Varargs: sumAll(1,2,3,4,5) = $sumVarargs")
        Text("Higher-order: operate(4,5,multiply) = $product")
        Text("Extension: \"Racecar\".isPalindrome() = $palindrome")
        Text("Tail-recursive: factorial(5) = $fact")
    }
}

@Composable
fun demonstrateClassesAndObjects() {
    // Usage examples
    val person = Person("Alice", 25)
    val user = User(1, "john_doe", "john@example.com")
    val   priority   =   Priority.HIGH   val
    config = AppConfig.getConfig() val
    area = MathUtils.circleArea(5.0)
    val dog = Dog("Buddy")
    val circle = Circle(10.0)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Class: ${person.introduce()}")
        Text("Data class: User(username=${user.username}, email=${user.email})")
        Text("Enum: ${priority.getDisplayName()} = ${priority.value}")
        Text("Object: $config")
        Text("Companion object: Circle area (r=5) = $area")
        Text("Abstract class: ${dog.makeSound()}")
        Text("Interface: ${circle.draw()}") }
}

@Composable
fun InteractiveExample(onResult: (String) -> Unit) { var
    input1 by remember { mutableStateOf("") } var input2
    by remember { mutableStateOf("") } var operation by
    remember { mutableStateOf("add") }

    Column(verticalArrangement = Arrangement.spacedBy(8.dp)) { Text("Try
        it yourself!")
```

```
OutlinedTextField( value
    = input1,
    onValueChange = { input1 = it }, label
    = { Text("Number 1") },
    modifier = Modifier.fillMaxWidth()
)

OutlinedTextField( value = input2,
    onValueChange = { input2 = it },
    label = { Text("Number 2") },
    modifier = Modifier.fillMaxWidth()
)
Row(                    horizontalArrangement                    =
    Arrangement.spacedBy(8.dp),          modifier          =
    Modifier.fillMaxWidth()
) {
    RadioButton(       selected       =
        operation == "add",
        onClick = { operation = "add" }
    )
    Text("Add")

    RadioButton( selected = operation ==
        "subtract",
        onClick = { operation = "subtract" }
    )
    Text("Subtract")

    RadioButton( selected = operation ==
        "multiply",
        onClick = { operation = "multiply" }
    )
    Text("Multiply")
}

Button( onClick
    = {
        val num1 = input1.toIntOrNull() ?: 0 val
        num2 = input2.toIntOrNull() ?: 0 val
        result = when (operation) {
            "add" -> num1 + num2
            "subtract" -> num1 - num2 "multiply"
            -> num1 * num2
            else -> 0
```

```
            }
            onResult("$num1 $operation $num2 = $result")
        },
        modifier = Modifier.align(Alignment.CenterHorizontally)
    ) {
        Text("Calculate")
    }
  }
}

@Preview(showBackground = true)
@Composable
fun KotlinFundamentalsDemoPreview() {
  Basic1Theme {
    KotlinFundamentalsDemo()
  }
}
```

**Output**

**Expt No : 4** **Date:**

## Build First Android App

**Aim**

To create and execute a basic Android application using Kotlin.

**Algorithm**

1. Open Android Studio.
2. Create new project → Empty Activity.
3. Design UI using XML layout.
4. Write Kotlin logic in Activity file.
5. Build the project.

6. Run app on emulator/device.

**AndroidManifest.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools">
```

```
<application android:allowBackup="true"
```

```
android:dataExtractionRules="@xml/data_extraction_rules
" android:fullBackupContent="@xml/backup_rules"
android:icon="@mipmap/ic_launcher"
android:label="@string/app_name"
android:supportsRtl="true"
android:theme="@style/Theme.GraphicalPrimitives"
tools:targetApi="31">
<activity
android:name=".MainActivity"
android:exported="true">
<intent-filter>
<action android:name="android.intent.action.MAIN" />
<category android:name="android.intent.category.LAUNCHER" />
</intent-filter>
</activity>
</application>
</manifest>
```

**activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.Constrai
ntLayout
xmlns:android="http://schemas.android.co
m/apk/res/android"
xmlns:app="http://schemas.android.com/ap
k/res-auto"
xmlns:tools="http://schemas.android.com/t
ools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
<org.rajalakshmi.graphicalprimitives.SampleCanvas
android:layout_width="match_parent"
android:layout_height="match_parent">
</org.rajalakshmi.graphicalprimitives.SampleCanvas>
</androidx.constraintlayout.widget.ConstraintLayout>
```

**MainActivity.kt**

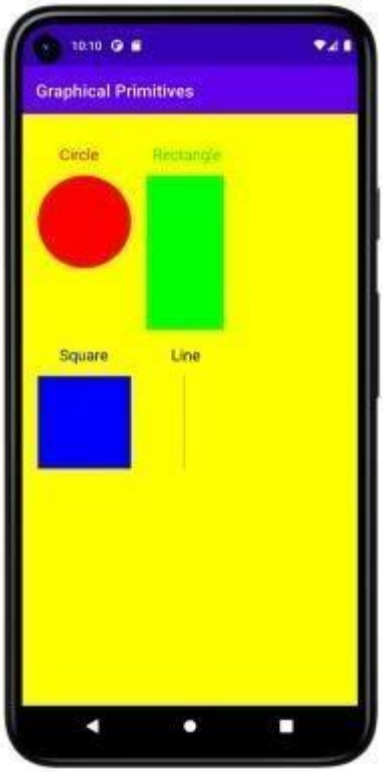
```
package org.rajalakshmi.graphicalprimitives import
androidx.appcompat.app.AppCompatActivity import
```

```
android.os.Bundle class MainActivity :
AppCompatActivity() { override fun
onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_main)
}
}
```

**SampleCanvas.kt**

```
package org.rajalakshmi.graphicalprimitives
import android.content.Context import
android.graphics.Canvas import
android.graphics.Color import
android.graphics.Paint import
android.util.AttributeSet import
android.view.View
class SampleCanvas @JvmOverloads constructor(
context: Context, attrs: AttributeSet? = null, defStyleAttr: Int = 0
) : View(context, attrs, defStyleAttr) {
override fun onDraw(canvas: Canvas?) {
super.onDraw(canvas)
val paint : Paint = Paint()
paint.setColor(Color.YELLOW)
canvas?.drawPaint(paint)
paint.setTextSize(50f);
paint.setColor(Color.RED);
canvas?.drawText("Circle", 120f, 150f, paint);
canvas?.drawCircle(200f, 350f, 150f, paint);
paint.setColor(Color.GREEN);

canvas?.drawText("Rectangle", 420f, 150f, paint); canvas?.drawRect(400f,
200f, 650f, 700f, paint); paint.setColor(Color.BLUE);
canvas?.drawText("Square", 120f, 800f, paint); canvas?.drawRect(50f,
850f, 350f, 1150f, paint); paint.setColor(Color.BLACK);
canvas?.drawText("Line", 480f, 800f, paint); canvas?.drawLine(520f, 850f,
520f, 1150f, paint);
}
}
```

**Output**




**Expt No : 5** **Date:**

## Layouts

**Aim**

To design user interfaces using Android layout managers.

**Algorithm**

1. Open XML layout file.
2. Select layout type.
3. Add UI components.
4. Set layout parameters.
5. Preview and run application.

Mainactivity.kt

```
package com.example.androidlayoutmanagers

import android.os.Bundle
import androidx.activity.ComponentActivity import
androidx.activity.compose.setContent import
androidx.activity.enableEdgeToEdge import
androidx.compose.foundation.layout.* import
androidx.compose.foundation.rememberScrollState import
androidx.compose.foundation.verticalScroll import
```

```
androidx.compose.material3.*                    import
androidx.compose.material3.ExperimentalMaterial3Api
import          androidx.compose.runtime.*      import
androidx.compose.ui.Alignment                   import
androidx.compose.ui.Modifier                    import
androidx.compose.ui.text.font.FontWeight        import
androidx.compose.ui.unit.dp                     import
androidx.compose.ui.unit.sp
import com.example.androidlayoutmanagers.ui.theme.AndroidLayoutManagersTheme

class MainActivity : ComponentActivity() { override
   fun onCreate(savedInstanceState: Bundle?) {
   super.onCreate(savedInstanceState)
      enableEdgeToEdge()
      setContent {
         AndroidLayoutManagersTheme {
            Surface(          modifier          =
               Modifier.fillMaxSize(),
               color = MaterialTheme.colorScheme.background
            ) {
               LayoutManagersApp()
            }
         }
      }
   }
}

@Composable          fun
LayoutManagersApp() {
   var selectedDemo by remember { mutableStateOf<DemoType?>(null) }

   selectedDemo?.let { demo ->
      DemoScreen(demo = demo, onBack = { selectedDemo = null })
   } ?: MainScreen(onDemoSelected = { selectedDemo = it })
}

@Composable
fun MainScreen(onDemoSelected: (DemoType) -> Unit) {
   Column(   modifier   =
      Modifier
         .fillMaxSize()
         .verticalScroll(rememberScrollState())
         .padding(16.dp),        verticalArrangement        =
      Arrangement.spacedBy(16.dp)
```

```
    ) {
        Text( text = "Android Layout
            Managers", fontSize = 32.sp,
            fontWeight = FontWeight.Bold,
            modifier = Modifier.align(Alignment.CenterHorizontally)
        )

        Text( text = "Explore different layout managers in Jetpack
            Compose", fontSize = 16.sp, color =
            MaterialTheme.colorScheme.onSurfaceVariant, modifier =
            Modifier.align(Alignment.CenterHorizontally)
        )

        Spacer(modifier = Modifier.height(16.dp))

        DemoType.values().forEach { demo ->
            Card( modifier = Modifier.fillMaxWidth(), elevation =
                CardDefaults.cardElevation(defaultElevation = 4.dp), onClick =
                { onDemoSelected(demo) }
            ) {
                Column( modifier =
                    Modifier
                    .fillMaxWidth()
                        .padding(16.dp)
                ) {
                    Text( text =
                        demo.title,
                        fontSize = 20.sp,
                        fontWeight = FontWeight.Bold
                    )
                    Spacer(modifier = Modifier.height(4.dp))
                    Text( text =
                        demo.description,
                        fontSize = 14.sp,
                        color = MaterialTheme.colorScheme.onSurfaceVariant
                    )
                }
            }
        }
    }
}

@OptIn(ExperimentalMaterial3Api::class)
@Composable
```

```
fun DemoScreen(demo: DemoType, onBack: () -> Unit) {
   Column(modifier = Modifier.fillMaxSize()) {
      TopAppBar(
         title = { Text(demo.title) }, navigationIcon
         = {
            IconButton(onClick = onBack) { Text("←
               Back")
            }
         }
      )

      Box(modifier = Modifier.fillMaxSize()) { when
         (demo) {
            DemoType.COLUMN -> ColumnExample()
            DemoType.ROW -> RowExample()
            DemoType.BOX -> BoxExample()
            DemoType.LAZY_COLUMN -> LazyColumnExample()
            DemoType.LAZY_ROW -> LazyRowExample()
            DemoType.MIXED_LAZY -> MixedLazyLayoutExample()
            DemoType.CONSTRAINT_BASIC -> ConstraintLayoutBasicExample()
            DemoType.CONSTRAINT_COMPLEX                                    ->
            ConstraintLayoutComplexExample()
            DemoType.CONSTRAINT_GUIDELINES ->
ConstraintLayoutGuidelinesExample()
            DemoType.GRID_FIXED          ->          LazyVerticalGridExample()
            DemoType.GRID_ADAPTIVE -> AdaptiveGridExample()
            DemoType.GRID_VARIABLE -> VariableGridExample()
            DemoType.PRODUCT_GRID -> ProductGridExample()
         }
      }
   }
}

enum class DemoType(
   val title: String, val
   description: String
) {
   COLUMN("Column Layout", "Vertical arrangement of components"),
   ROW("Row Layout", "Horizontal arrangement of components"),
   BOX("Box Layout", "Stacked components with positioning"),
   LAZY_COLUMN("LazyColumn", "Efficient vertical scrolling lists"),
   LAZY_ROW("LazyRow", "Efficient horizontal scrolling lists"),
   MIXED_LAZY("Mixed Lazy Layout", "Combination of LazyColumn and LazyRow"),
```

```
    CONSTRAINT_BASIC("ConstraintLayout Basic", "Basic constraint positioning"),
    CONSTRAINT_COMPLEX("ConstraintLayout Complex", "Complex profile layout"),
    CONSTRAINT_GUIDELINES("ConstraintLayout Guidelines", "Layout with
guidelines"),
    GRID_FIXED("Fixed Grid", "Fixed number of columns"),
    GRID_ADAPTIVE("Adaptive Grid", "Adaptive column sizing"),
    GRID_VARIABLE("Variable Grid", "Variable item sizes"),
    PRODUCT_GRID("Product Grid", "Real-world product grid")
}
```

**settings.gradle.kts**

```
pluginManagement { repositories { google { content
    {        includeGroupByRegex("com\\.android.*")
    includeGroupByRegex("com\\.google.*")
             includeGroupByRegex("androidx.*")
          }
       }
       mavenCentral()
       gradlePluginPortal()
    }
}
```

```
dependencyResolutionManagement {
    repositoriesMode.set(RepositoriesMode.FAIL_ON_PROJECT_REPOS
    ) repositories { google() mavenCentral()
    }
}

rootProject.name = "Android layout managers"
include(":app")
```

**ConstraintsLayoutExamples.kts**

```
package com.example.androidlayoutmanagers import
androidx.compose.foundation.background import
androidx.compose.foundation.layout.* import
androidx.compose.foundation.shape.RoundedCornerShape
import androidx.compose.material3.* import
androidx.compose.runtime.* import
androidx.compose.ui.Alignment import
androidx.compose.ui.Modifier import
androidx.compose.ui.graphics.Color import
androidx.compose.ui.text.font.FontWeight import
androidx.compose.ui.unit.dp import androidx.compose.ui.unit.sp
import androidx.constraintlayout.compose.ConstraintLayout import
androidx.constraintlayout.compose.ConstraintSet
import androidx.constraintlayout.compose.Dimension

@Composable
fun ConstraintLayoutBasicExample() {
   ConstraintLayout(
      modifier = Modifier
         .fillMaxSize()
         .padding(16.dp)
   ) { val (title, subtitle, button, image) = createRefs()

      Text( text = "ConstraintLayout Basic",
         fontSize = 24.sp, fontWeight =
         FontWeight.Bold, modifier =
         Modifier.constrainAs(title) {
         top.linkTo(parent.top, margin = 16.dp)
         start.linkTo(parent.start)
            end.linkTo(parent.end)
         }
      )

      Text( text = "Positioning elements with constraints",
         fontSize = 16.sp, color =
         MaterialTheme.colorScheme.onSurfaceVariant,
         modifier = Modifier.constrainAs(subtitle) {
            top.linkTo(title.bottom, margin = 8.dp)
            start.linkTo(parent.start)
            end.linkTo(parent.end)
         }
      )
```

```
        Button( onClick
            = { },
            modifier = Modifier.constrainAs(button) {
                bottom.linkTo(parent.bottom, margin = 32.dp)
                start.linkTo(parent.start)
                end.linkTo(parent.end)
            }
        ) {
            Text("Centered Button")
        }

        Box(      modifier      =
            Modifier
                .size(100.dp)
                .background(
                    Color.Blue,
                    RoundedCornerShape(12.dp)
                )
                .constrainAs(image)                              {
                    top.linkTo(subtitle.bottom, margin = 24.dp)
                    bottom.linkTo(button.top, margin = 24.dp)
                    start.linkTo(parent.start)
                    end.linkTo(parent.end)
                },
            contentAlignment = Alignment.Center
        ) {
            Text( text = "Center
                Box", color =
                Color.White,
                fontWeight = FontWeight.Bold
            )
        }
    }
}

@Composable
fun ConstraintLayoutComplexExample() {
    ConstraintLayout(
        modifier = Modifier
            .fillMaxSize()
            .padding(16.dp)
    ) {
```

```
val (header, profilePic, userName, userEmail, statsContainer,
bioText, actionButton, settingsButton) = createRefs() val barrier
= createEndBarrier(profilePic, userName, userEmail)

Text( text = "Profile Layout", fontSize =
   20.sp, fontWeight = FontWeight.Bold,
   modifier = Modifier.constrainAs(header) {
      top.linkTo(parent.top)
      start.linkTo(parent.start)
      end.linkTo(parent.end)
   }
)

Box(     modifier     =
   Modifier
      .size(80.dp)
      .background(
         Color.Green,
         RoundedCornerShape(40.dp)
      )
      .constrainAs(profilePic) { top.linkTo(header.bottom,
         margin = 16.dp)
         start.linkTo(parent.start)
      },
   contentAlignment = Alignment.Center
) {
   Text( text = "PIC",
      color = Color.White,
      fontWeight = FontWeight.Bold
   )
}

Text( text = "John
   Doe", fontSize =
   18.sp, fontWeight
   =
   FontWeight.Bold,
   modifier = Modifier.constrainAs(userName) {
      start.linkTo(profilePic.end, margin = 16.dp)
      top.linkTo(profilePic.top)
   }
)
```

```
Text( text =
    "john.doe@example.com",
    fontSize = 14.sp, color =
    MaterialTheme.colorScheme.onSurfaceVariant,
    modifier = Modifier.constrainAs(userEmail) {
    start.linkTo(profilePic.end, margin = 16.dp)
    top.linkTo(userName.bottom, margin = 4.dp)
    }
)

Box( modifier =
    Modifier
        .constrainAs(statsContainer) { top.linkTo(profilePic.bottom,
            margin = 16.dp)
            start.linkTo(parent.start)
            end.linkTo(parent.end)
            width = Dimension.fillToConstraints
        } )
{
    Row( modifier =
        Modifier.fillMaxWidth(),
        horizontalArrangement = Arrangement.SpaceEvenly
    ) {
        StatItem("Posts", "150", Modifier)
        StatItem("Followers", "2.5K", Modifier)
        StatItem("Following", "500", Modifier)
    }
}

Text( text = "Passionate developer and Android enthusiast. Love creating beautiful
    and
functional mobile applications.",
    fontSize = 14.sp,
    modifier = Modifier.constrainAs(bioText) {
        top.linkTo(statsContainer.bottom, margin = 16.dp)
        start.linkTo(parent.start) end.linkTo(parent.end)
        width = Dimension.fillToConstraints
    }
)
Button( onClick
    = { },
    modifier = Modifier.constrainAs(actionButton) {
```

```
            top.linkTo(bioText.bottom,  margin  =  16.dp)
            start.linkTo(parent.start)
            end.linkTo(settingsButton.start, margin = 8.dp)
            width = Dimension.fillToConstraints
        }
    ) {
        Text("Follow")
    }

    OutlinedButton(
        onClick = { },
        modifier = Modifier.constrainAs(settingsButton) {
            top.linkTo(bioText.bottom, margin = 16.dp) end.linkTo(parent.end)
            width = Dimension.value(120.dp)
        }
    ) {
        Text("Message")
    }
  }
}

@Composable
private fun StatItem(label: String, value: String, modifier: Modifier) {
  Column(                    horizontalAlignment                    =
    Alignment.CenterHorizontally, modifier = modifier
  ) {
    Text( text = value,
        fontSize = 18.sp,
        fontWeight = FontWeight.Bold
    )
    Text( text = label,
        fontSize = 12.sp,
        color = MaterialTheme.colorScheme.onSurfaceVariant
    )
  }
}

@Composable
fun ConstraintLayoutGuidelinesExample() {
  ConstraintLayout(
    modifier = Modifier
        .fillMaxSize()
        .padding(16.dp)
```

```
) {
    val (title, leftCard, centerCard, rightCard, bottomBar) = createRefs()

    val verticalGuideline = createGuidelineFromStart(0.33f)
    val verticalGuideline2 = createGuidelineFromEnd(0.33f)
    val horizontalGuideline = createGuidelineFromTop(0.4f)

    Text( text = "Guidelines Layout", fontSize
        = 20.sp, fontWeight =
        FontWeight.Bold, modifier =
        Modifier.constrainAs(title) {
        top.linkTo(parent.top)
        start.linkTo(parent.start)
            end.linkTo(parent.end)
        }
    )

    Card( modifier =
        Modifier
            .fillMaxHeight()
            .constrainAs(leftCard) { top.linkTo(title.bottom,
                margin = 16.dp) bottom.linkTo(bottomBar.top,
                margin = 16.dp)
                start.linkTo(parent.start)
                end.linkTo(verticalGuideline)
            },
        colors = CardDefaults.cardColors(
            containerColor = Color.Red.copy(alpha = 0.7f)
        )
    ) {
        Box( contentAlignment =
            Alignment.Center,
            modifier = Modifier.fillMaxSize()
        ) {
            Text( text =
                "Left\n33%", color
                = Color.White,
                fontWeight = FontWeight.Bold
            )
        }
    }
```

```
Card( modifier =
    Modifier
    .fillMaxHeight()
        .constrainAs(centerCard)                    {
            top.linkTo(title.bottom,    margin    =    16.dp)
            bottom.linkTo(bottomBar.top, margin = 16.dp)
            start.linkTo(verticalGuideline)
            end.linkTo(verticalGuideline2)
        },
    colors = CardDefaults.cardColors(
        containerColor = Color.Green.copy(alpha = 0.7f)
    )
) {
    Box(            contentAlignment            =
        Alignment.Center,
        modifier = Modifier.fillMaxSize()
    ) {
        Text(          text          =
            "Center\n34%",  color
            = Color.White,
            fontWeight = FontWeight.Bold
        )
    }
}

Card(     modifier     =
    Modifier
        .fillMaxHeight()
        .constrainAs(rightCard) { top.linkTo(title.bottom,
            margin = 16.dp) bottom.linkTo(bottomBar.top,
            margin = 16.dp)
            start.linkTo(verticalGuideline2)
            end.linkTo(parent.end)
        },
    colors = CardDefaults.cardColors(
        containerColor = Color.Blue.copy(alpha = 0.7f)
    )
) {
    Box(            contentAlignment            =
        Alignment.Center,
        modifier = Modifier.fillMaxSize()
    ) {
```

```
            Text(        text        =
                "Right\n33%", color
                = Color.White,
                fontWeight = FontWeight.Bold
            )
        }
    }
    Card(    modifier    =
        Modifier
            .height(60.dp)
            .constrainAs(bottomBar) {
                bottom.linkTo(parent.bottom)
                start.linkTo(parent.start)
                end.linkTo(parent.end)
            },
        colors = CardDefaults.cardColors(
            containerColor = MaterialTheme.colorScheme.surfaceVariant
        )
    ) {
        Box(            contentAlignment            =
            Alignment.Center,
            modifier = Modifier.fillMaxSize()
        ) {
            Text( text = "Bottom Navigation
                Bar",
                fontWeight = FontWeight.Medium
            )
        }
    }
  }
}
```

**GridLayoutExamples.kt**

```
package        com.example.androidlayoutmanagers        import
androidx.compose.foundation.background                   import
androidx.compose.foundation.layout.*                     import
androidx.compose.foundation.lazy.grid.GridCells          import
androidx.compose.foundation.lazy.grid.LazyVerticalGrid import
androidx.compose.foundation.lazy.grid.items
import androidx.compose.foundation.lazy.grid.rememberLazyGridState
import        androidx.compose.foundation.shape.RoundedCornerShape
import                androidx.compose.material3.*                import
```

```
androidx.compose.runtime.*   import   androidx.compose.ui.Alignment
import                androidx.compose.ui.Modifier                import
androidx.compose.ui.graphics.Color                                 import
androidx.compose.ui.text.font.FontWeight                           import
androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp

data class GridItem(val id: Int, val title: String, val color: Color)

@Composable
fun LazyVerticalGridExample() { val
   items = remember {
      (1..50).map { index -> val
         colors = listOf(
            Color.Red, Color.Blue, Color.Green, Color.Yellow,
            Color.Magenta, Color.Cyan, Color.Gray, Color(0xFFFF6B35)
         )
         GridItem( id = index,
            title = "Item
            $index",
            color = colors[index % colors.size]
         )
      }
   }

   val gridState = rememberLazyGridState()

   Column(          modifier          =
      Modifier.fillMaxSize()
   ) {
      Text(  text  =  "LazyVerticalGrid  -  2
         Columns",     fontSize     =     24.sp,
         fontWeight = FontWeight.Bold,
         modifier = Modifier.padding(16.dp)
      )

      LazyVerticalGrid(     columns     =
         GridCells.Fixed(2),   modifier   =
         Modifier.fillMaxSize(),
         state = gridState,
         contentPadding          =          PaddingValues(16.dp),
         horizontalArrangement = Arrangement.spacedBy(12.dp),
         verticalArrangement = Arrangement.spacedBy(12.dp)
      ) {
```

```
        items(items) { item ->
            Card(    modifier    =
                Modifier
                    .fillMaxWidth()
                    .aspectRatio(1f),                    elevation                    =
                CardDefaults.cardElevation(defaultElevation = 4.dp)
            ) {
                Box(    modifier    =
                    Modifier
                        .fillMaxSize()
                        .background(item.color.copy(alpha = 0.8f)), contentAlignment
                    = Alignment.Center
                ) {
                    Text( text = item.title, color =
                        Color.White,    fontWeight    =
                        FontWeight.Bold,
                        fontSize = 16.sp
                    )
                }
            }
        }
    }
  }
}

@Composable                fun
AdaptiveGridExample() { val
items = remember {
    (1..30).map { index -> val
        colors = listOf(
            Color(0xFF6200EE), Color(0xFF03DAC6), Color(0xFFCF6679),
            Color(0xFF3700B3), Color(0xFF018786), Color(0xFFBB86FC)
        )
        GridItem( id
            = index,
            title = "Adaptive $index", color =
            colors[index % colors.size]
        )
    }
  }

  Column(          modifier          =
    Modifier.fillMaxSize()
```

```
) {
    Text( text = "Adaptive Grid - Min
        120dp", fontSize = 24.sp,
        fontWeight = FontWeight.Bold,
        modifier = Modifier.padding(16.dp)
    )

    LazyVerticalGrid( columns =
        GridCells.Adaptive(minSize = 120.dp),
        modifier = Modifier.fillMaxSize(), contentPadding =
        PaddingValues(16.dp), horizontalArrangement =
        Arrangement.spacedBy(8.dp), verticalArrangement =
        Arrangement.spacedBy(8.dp)
    ) {
        items(items) { item ->
            Card(
                modifier = Modifier
                    .fillMaxWidth()
                    .height(100.dp), elevation =
                CardDefaults.cardElevation(defaultElevation = 3.dp)
            ) {
                Box(
```

**LazyLayoutExamples.kt**

```
package com.example.androidlayoutmanagers

import androidx.compose.foundation.background
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.lazy.LazyColumn
import androidx.compose.foundation.lazy.LazyRow
import androidx.compose.foundation.lazy.items
import androidx.compose.foundation.lazy.rememberLazyListState
import androidx.compose.foundation.shape.RoundedCornerShape
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.graphics.Color
import androidx.compose.ui.text.font.FontWeight
import
```

```
androidx.compose.ui.unit.dp  import  androidx.compose.ui.unit.sp
data class Item(val id: Int, val title: String, val description: String)

@Composable                fun
LazyColumnExample() { val
items = remember {
      (1..50).map { index ->
         Item( id = index, title =
            "Item $index",
            description = "This is the description for item number $index"
         )
      }
   }

   val listState = rememberLazyListState()

   Column(          modifier          =
      Modifier.fillMaxSize()
   ) {
      Text( text = "LazyColumn - ${items.size}
         items", fontSize = 24.sp, fontWeight =
         FontWeight.Bold,
         modifier = Modifier.padding(16.dp)
      )

      LazyColumn(        modifier        =
         Modifier.fillMaxSize(),
         state = listState,
         contentPadding = PaddingValues(16.dp), verticalArrangement
         = Arrangement.spacedBy(8.dp)
      ) {
         items(items) { item ->
            Card(    modifier    =
               Modifier
                  .fillMaxWidth(),                    elevation                    =
               CardDefaults.cardElevation(defaultElevation = 4.dp)
            ) {
               Column( modifier =
                  Modifier
                     .fillMaxWidth()
                     .padding(16.dp)
               ) {
```

```
                Text( text = item.title, fontSize =
                    18.sp,          fontWeight          =
                    FontWeight.Bold,
                    color = MaterialTheme.colorScheme.primary
                )
                Spacer(modifier = Modifier.height(4.dp))
                Text(           text           =
                    item.description,
                    fontSize = 14.sp,
                    color = MaterialTheme.colorScheme.onSurfaceVariant
                )
            }
        }
    }
  }
}
}

@Composable fun
LazyRowExample() { val
categories = remember {
listOf(
        "Technology", "Sports", "Music", "Movies", "Books",
        "Travel", "Food", "Fashion", "Gaming", "Art",
        "Science", "History", "Nature", "Photography", "Dance"
    )
  }

  val listState = rememberLazyListState()

  Column(   modifier   =
    Modifier
      .fillMaxSize()
      .padding(16.dp)
  ) {
    Text(   text   =   "LazyRow   -   Horizontal
      Scrolling", fontSize = 24.sp, fontWeight
      = FontWeight.Bold,
      modifier = Modifier.padding(bottom = 16.dp)
    )

    LazyRow(
      state = listState,
```

```
    horizontalArrangement = Arrangement.spacedBy(12.dp), contentPadding
    = PaddingValues(horizontal = 16.dp)
) {
    items(categories) { category ->
        Card(     modifier     =
            Modifier
                .width(120.dp)
                .height(80.dp), shape =
            RoundedCornerShape(12.dp),
            elevation = CardDefaults.cardElevation(defaultElevation = 6.dp)
        ) {
            Box(     modifier     =
                Modifier
                    .fillMaxSize()
                    .background(
                        Color( red = (0.3f + (category.hashCode() % 100) /
                            200f), green = (0.5f + (category.hashCode() % 150)
                            / 300f), blue = (0.7f + (category.hashCode() % 200)
                            / 400f)
                        )
                    ),
                contentAlignment = Alignment.Center
            ) {
                Text(  text  =  category,
                    color = Color.White,
                    fontWeight          =
                    FontWeight.Bold,
                    modifier = Modifier.padding(8.dp)
                )
            }
        }
    }
}

Spacer(modifier = Modifier.height(24.dp))

Text( text = "Featured Items",
    fontSize = 20.sp, fontWeight =
    FontWeight.Bold,
    modifier = Modifier.padding(bottom = 12.dp)
)

val featuredItems = remember {
    (1..20).map { index ->
```

```
            "Featured $index"
        }
    }

    LazyRow(horizontalArrangement = Arrangement.spacedBy(8.dp), contentPadding = PaddingValues(horizontal = 16.dp)
    ) {
        items(featuredItems) { item ->
            Card(modifier = Modifier
                .width(150.dp)
                .height(100.dp), colors = CardDefaults.cardColors(
                    containerColor = MaterialTheme.colorScheme.surfaceVariant
                )
            ) {
                Box(contentAlignment = Alignment.Center
                ) {
                    Text(text = item,
                        fontWeight = FontWeight.Medium,
                        modifier = Modifier.padding(12.dp)
                    )
                }
            }
        }
    }
  }
}

@Composable
fun MixedLazyLayoutExample() {
  val sections = remember {
  listOf(
        "Header" to listOf("Welcome to Lazy Layouts"),
        "Categories" to listOf("Tech", "Sports", "Music", "Movies", "Books"),
        "Popular Items" to (1..10).map { "Popular Item $it" },
        "Recent Items" to (1..15).map { "Recent Item $it" }
    )
  }
```

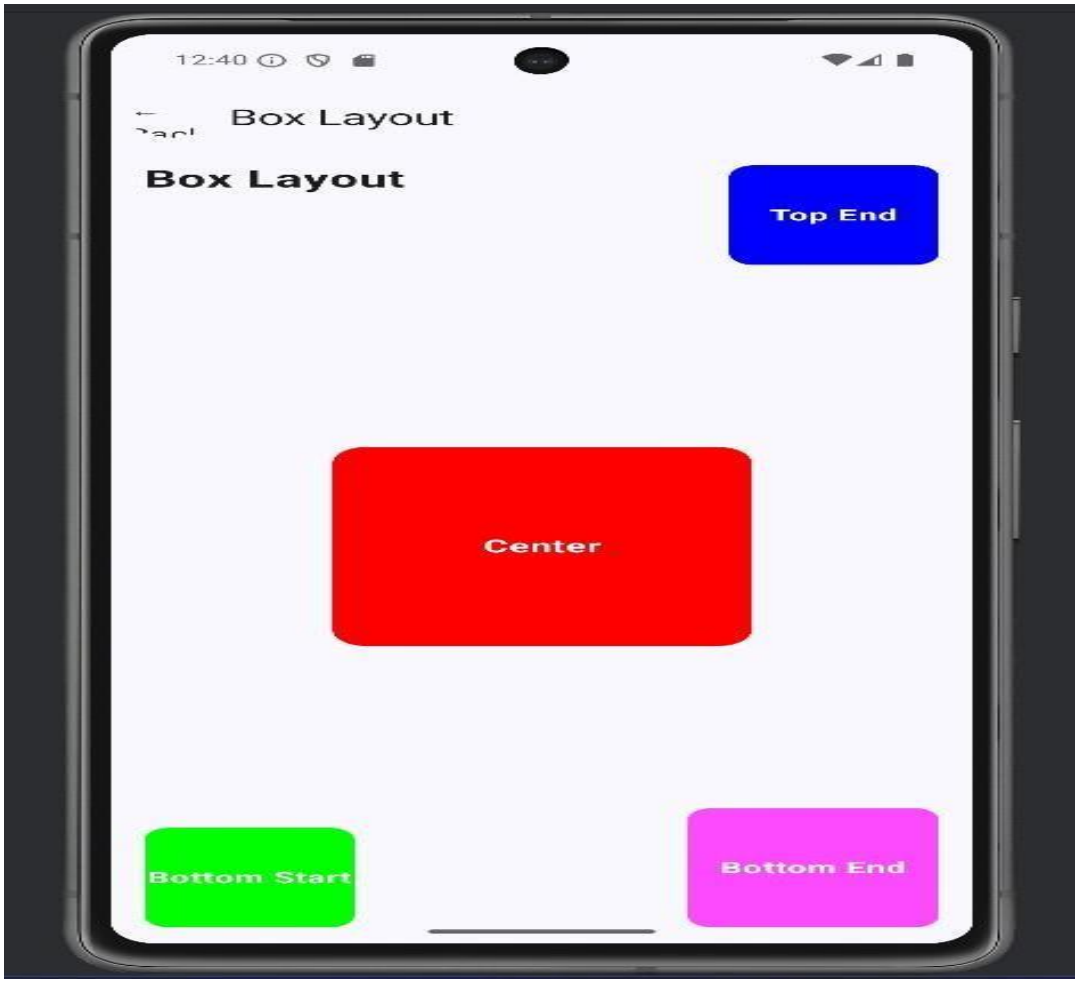
```
LazyColumn(                modifier              =
    Modifier.fillMaxSize(), contentPadding =
    PaddingValues(16.dp),
    verticalArrangement = Arrangement.spacedBy(16.dp)
) {
    sections.forEach { (sectionTitle, items) -> item
        {
            Text( text = sectionTitle, fontSize
                =     20.sp,     fontWeight     =
                FontWeight.Bold,
                modifier = Modifier.padding(vertical = 8.dp)
            )
        }

        if (sectionTitle == "Categories") { item
            {
                LazyRow(                horizontalArrangement                =
                    Arrangement.spacedBy(8.dp)
                ) {
                    items(items) { category ->
                        Card(     modifier     =
                            Modifier
                                .width(100.dp)
                                .height(60.dp),        colors        =        CardDefaults.cardColors(
                            containerColor = MaterialTheme.colorScheme.primaryContainer
                            )
                        ) {
                            Box(            contentAlignment            =
                                Alignment.Center
                            ) {
                                Text( text =
                                    category,
                                    fontWeight =
```

```
                    FontWeight.M
                    edium,
                   color = MaterialTheme.colorScheme.onPrimaryContainer
                )
            }
        }
    }
  }
}
```

```
else { items(items) {
  item ->
    Card( modifier =
      Modifier.fillMaxWidth(),
     elevation = CardDefaults.cardElevation(defaultElevation = 2.dp)
```

```
                ) {
                    Text(   text   =
                        item,
                        modifier = Modifier.padding(16.dp)
                    )
                }
            }
        }
    }
}
```

**Output**


12:40
Box Layout
Box Layout
Top End
Center
Bottom Start
Bottom End

**Expt No : 6** **Date:**

## App Navigation

**Aim**

To implement navigation between multiple screens in Android app.

**Algorithm**

1. Create multiple activities/fragments.
2. Use Intent to navigate. 3. Pass data using extras.
4. Implement Navigation component.
5. Test navigation flow.

**CODING**

**MainActivity.kt**

```
package com.example.sai1 import
android.content.Intent import
androidx.appcompat.app.AppCompatActivity
import android.os.Bundle import
android.widget.Button

class MainActivity : AppCompatActivity() {
   override fun onCreate(savedInstanceState: Bundle?) {
      super.onCreate(savedInstanceState)
      setContentView(R.layout.activity_main)

      val btnPage1: Button = findViewById(R.id.btnPage1) val
      btnPage2: Button = findViewById(R.id.btnPage2) val
      btnPage3: Button = findViewById(R.id.btnPage3) val
      btnPage4: Button = findViewById(R.id.btnPage4)

      btnPage1.setOnClickListener {
         startActivity(Intent(this, Page1Activity::class.java))
      }

      btnPage2.setOnClickListener {
         startActivity(Intent(this, Page2Activity::class.java))
      }

      btnPage3.setOnClickListener {
```

```
        startActivity(Intent(this,  Page3Activity::class.java))
    }
    btnPage4.setOnClickListener {
    startActivity(Intent(this, Page4Activity::class.java)) }
  }
}
```

**Activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent" android:orientation="vertical"
  android:gravity="center" android:padding="16dp"
  android:background="@color/white">

  <TextView
    android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content
        " android:text="GUI Components"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="40dp" />

    <Button android:id="@+id/btnPage1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content
        " android:text="Page 1 - Initial State"
        android:textSize="16sp"
        android:layout_marginBottom="16dp" />

    <Button android:id="@+id/btnPage2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content
        " android:text="Page 2 - Large Font"
        android:textSize="16sp"
        android:layout_marginBottom="16dp" />

    <Button android:id="@+id/btnPage3"
        android:layout_width="match_parent"
        android:layout_height="wrap_content
        " android:text="Page 3 - Green Text"
        android:textSize="16sp"
        android:layout_marginBottom="16dp" />

    <Button          android:id="@+id/btnPage4"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Page 4 - Blue Background"
        android:textSize="16sp" />

</LinearLayout>
```

**AndroidManifest.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<manifest                    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
```

```
<application android:allowBackup="true"
```

```
        android:dataExtractionRules="@xml/data_extraction_rules
        " android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:supportsRtl="true"
        android:theme="@style/Theme.Sai1"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <activity
            android:name=".Page1Activity"
            android:exported="false" />
        <activity
            android:name=".Page2Activity"
            android:exported="false" />
        <activity
            android:name=".Page3Activity"
            android:exported="false" />
        <activity
            android:name=".Page4Activity"
            android:exported="false" />
    </application>
</manifest>
```

## OUTPUT

**Expt No : 7** **Date:**

**Activity and Fragment Lifecycles**

**Aim**

To understand lifecycle methods of Activity and Fragment.

**Algorithm**

1. Create activity/fragment.
2. Override lifecycle methods.
3. Use Logcat for logging.
4. Rotate device and observe lifecycle.
5. Analyze lifecycle behavior.

**activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent" android:orientation="vertical"
tools:context=".MainActivity">
<TextView android:id="@+id/tvTitle"
android:layout_width="match_parent"
```

```
android:layout_height="wrap_content
" android:text="Student Details"
android:textAlignment="center"
android:textSize="24sp" />
<LinearLayout
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical">
<fragment
android:id="@+id/fragmentBasic"
android:name="org.rajalakshmi.androidfragments.StudentBasicDetails
" android:layout_width="match_parent"
android:layout_height="300dp" />
<fragment
android:id="@+id/fragmentMark"

android:name="org.rajalakshmi.androidfragments.StudentMarkDetails
" android:layout_width="match_parent"
android:layout_height="300dp" />
</LinearLayout>
</LinearLayout>
```

**fragment_student_basic_details.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".StudentBasicDetails">
<TextView
android:id="@+id/tvBasicDetails"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Basic Details"
android:textAlignment="center"
android:textSize="24sp" />
<TextView
android:id="@+id/tvRegisterNumber"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="50dp"
android:text="Register No." />
```

```
<EditText
android:id="@+id/etRegisterNumber"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="50dp"
android:ems="10"
android:hint="Register Number"
android:inputType="textPersonName" />
<TextView
android:id="@+id/tvName"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="125dp"
android:text="Name" />
<EditText
android:id="@+id/etName"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="125dp"
android:ems="10"
android:hint="Name"
android:inputType="textPersonName" />
<TextView
android:id="@+id/tvDepartment"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="200dp"
android:text="Department" />
<EditText
android:id="@+id/etDepartment"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="200dp"
android:ems="10"
android:hint="Department"
android:inputType="textPersonName" />
</FrameLayout>
```

**fragment_student_mark_details.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout
xmlns:android="http://schemas.android.com/apk/res/android"
```

```
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".StudentMarkDetails">
<TextView
android:id="@+id/tvBasicDetails"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Mark Details"
android:textAlignment="center"
android:textSize="24sp" />
<TextView
android:id="@+id/tvSSLC"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="50dp"
android:text="S.S.L.C." />
<EditText
android:id="@+id/etSSLC"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="50dp"
android:ems="10"
android:hint="S.S.L.C.                    Mark"
android:inputType="textPersonName" />
<TextView
android:id="@+id/tvHSc"
android:layout_width="wrap_content
"
android:layout_height="wrap_conten
t"
android:layout_marginTop="125dp"
android:text="H.Sc." />
<EditText
android:id="@+id/etHSC"
android:layout_width="wrap_content"
android:layout_height="wrap_content
" android:layout_marginLeft="150dp"
android:layout_marginTop="125dp"
android:ems="10"
android:hint="H.Sc. Mark"
android:inputType="textPersonName" />
<TextView android:id="@+id/tvUG"
android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content
" android:layout_marginTop="200dp"
android:text="U.G." />
<EditText
```

```
android:id="@+id/etUG"
android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content
" android:layout_marginLeft="150dp"
android:layout_marginTop="200dp"
android:ems="10" android:hint="U.G.
C.G.P.A."
android:inputType="textPersonName" />
</FrameLayout>
```

**MainActivity.kt**

```
package org.rajalakshmi.androidfragments import
androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
class MainActivity : AppCompatActivity() { override
fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_main)
}
}
```



StudentBasicDetails.kt

```
package org.rajalakshmi.androidfragments
import android.os.Bundle import
androidx.fragment.app.Fragment import
android.view.LayoutInflater import
android.view.View
import android.view.ViewGroup
// TODO: Rename parameter arguments, choose names that match //
the fragment initialization parameters, e.g. ARG_ITEM_NUMBER
private const val ARG_PARAM1 = "param1"
private const val ARG_PARAM2 = "param2"
/**
* A simple [Fragment] subclass.
* Use the [StudentBasicDetails.newInstance] factory method to * create an
  instance of this fragment.
*/
class StudentBasicDetails : Fragment() { // TODO:
Rename and change types of parameters
private var param1: String? = null private
var param2: String? = null
override fun onCreate(savedInstanceState: Bundle?)
{ super.onCreate(savedInstanceState) arguments?.let
{
param1 = it.getString(ARG_PARAM1)
```

```
param2 = it.getString(ARG_PARAM2)
}
}
override fun onCreateView(
inflater: LayoutInflater, container: ViewGroup?, savedInstanceState:
Bundle?
): View? {
// Inflate the layout for this fragment
return inflater.inflate(R.layout.fragment_student_basic_details, container,
false)
}
companion object {
/**
* Use this factory method to create a new instance of * this fragment using the provided
  parameters.
*
* @param param1 Parameter 1.
* @param param2 Parameter 2.
* @return A new instance of fragment StudentBasicDetails.
*/
// TODO: Rename and change types and number of parameters
@JvmStatic
fun newInstance(param1: String, param2: String) =
StudentBasicDetails().apply { arguments =
Bundle().apply { putString(ARG_PARAM1,
param1)
putString(ARG_PARAM2, param2)
}
}
}
}
```



StudentMarkDetails.kt

```
package org.rajalakshmi.androidfragments
import android.os.Bundle import
androidx.fragment.app.Fragment import
android.view.LayoutInflater import
android.view.View import
android.view.ViewGroup
// TODO: Rename parameter arguments, choose names that match //
the fragment initialization parameters, e.g. ARG_ITEM_NUMBER
private const val ARG_PARAM1 = "param1"
```

```
private const val ARG_PARAM2 = "param2"
/**
* A simple [Fragment] subclass.
* Use the [StudentMarkDetails.newInstance] factory method to * create an
  instance of this fragment.
*/
class StudentMarkDetails : Fragment() { // TODO:
Rename and change types of parameters
private var param1: String? = null private
var param2: String? = null
override fun onCreate(savedInstanceState: Bundle?)
{ super.onCreate(savedInstanceState) arguments?.let
{
param1 = it.getString(ARG_PARAM1)
param2 = it.getString(ARG_PARAM2)
}
}
override fun onCreateView(
inflater: LayoutInflater, container: ViewGroup?, savedInstanceState:
Bundle?
): View? {
// Inflate the layout for this fragment
return inflater.inflate(R.layout.fragment_student_mark_details, container,
false)
}
companion object {
/**
* Use this factory method to create a new instance of * this fragment using the provided
  parameters.
*
* @param param1 Parameter 1.
* @param param2 Parameter 2.
* @return A new instance of fragment StudentMarkDetails.
*/
// TODO: Rename and change types and number of parameters
@JvmStatic
fun newInstance(param1: String, param2: String) =
StudentMarkDetails().apply { arguments =
Bundle().apply { putString(ARG_PARAM1,
param1)
putString(ARG_PARAM2, param2)
}
}
```

}
}

**Output**

**Output**

**Expt No : 8** **Date:**

**App Architecture (UI Layer)**

**Aim**

To implement UI layer using MVVM architecture.

**Algorithm**

1. Create ViewModel class.
2. Add LiveData objects.
3. Bind UI with ViewModel.
4. Observe data changes.
5. Update UI automatically.

**Coding**

**Mainactivity.kt**

package com.example.apparchitectureuilayer

```
import android.os.Bundle
import androidx.activity.ComponentActivity import
androidx.activity.compose.setContent import androidx.activity.enableEdgeToEdge
import
com.example.apparchitectureuilayer.presentation.navigation.AppNavigationSimple
import com.example.apparchitectureuilayer.ui.theme.AppArchitectureUILayerTheme

class MainActivity : ComponentActivity() { override
   fun onCreate(savedInstanceState: Bundle?) {
   super.onCreate(savedInstanceState)
      enableEdgeToEdge()
      setContent {
         AppArchitectureUILayerTheme {
            AppNavigationSimple()
         }
      }
   }
}
```

**UserDetailScreen.kt** package

```
com.example.apparchitectureuilayer.presentation.screen

import androidx.compose.foundation.background
import androidx.compose.foundation.layout.* import
androidx.compose.foundation.lazy.LazyColumn import
androidx.compose.foundation.shape.CircleShape import
androidx.compose.material3.*
import androidx.compose.runtime.* import androidx.compose.ui.Alignment import
androidx.compose.ui.Modifier import androidx.compose.ui.draw.clip import
androidx.compose.ui.layout.ContentScale import
androidx.compose.ui.text.font.FontWeight import androidx.compose.ui.unit.dp
import androidx.hilt.navigation.compose.hiltViewModel import
coil.compose.AsyncImage import
com.example.apparchitectureuilayer.data.model.ApiResponse import
```

com.example.apparchitectureuilayer.presentation.viewmodel.UserDetailViewMod
el

```
@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun
    UserDetailScreen(
    userId:         String,
    onBack: () -> Unit,
    viewModel: UserDetailViewModel = hiltViewModel()
) {
    LaunchedEffect(userId)                                  {
        viewModel.loadUser(userId)
```

```
}                val                user                by

viewModel.user.collectAsState()

Scaffold( topBar
   = {
      TopAppBar(
         title = { Text("User Details") }, navigationIcon
         = {
            IconButton(onClick = onBack) { Text("←")
            }
         }
      )
   }
) { padding ->
   Box(     modifier     =
      Modifier
         .fillMaxSize()
         .padding(padding)
   ) {
      when        (user)        {        is
         ApiResponse.Loading -> {
            CircularProgressIndicator(        modifier        =
               Modifier.align(Alignment.Center)
            )
         }
         is ApiResponse.Success -> { val userData = (user as
            ApiResponse.Success).data
            LazyColumn(              modifier              =
               Modifier.fillMaxSize(), contentPadding =
               PaddingValues(16.dp),
               verticalArrangement = Arrangement.spacedBy(16.dp)
            ) {  item
               {
                  Row(    modifier    =    Modifier.fillMaxWidth(),
                     horizontalArrangement = Arrangement.Center
                  ) { if (userData.avatar != null)
                     {
                        AsyncImage(            model            =
                           userData.avatar,  contentDescription
                           =    "User    Avatar",    modifier    =
                           Modifier
                              .size(120.dp)
```

```
            .clip(CircleShape),
        contentScale = ContentScale.Crop
    )
} else {
    Box(    modifier    =
        Modifier
            .size(120.dp)
            .clip(CircleShape)
```

```
                .background(MaterialTheme.colorScheme.primaryContainer),
            contentAlignment = Alignment.Center
        ) {
            Text( text = userData.name.first().uppercase(), style =
                MaterialTheme.typography.headlineLarge, color =
                MaterialTheme.colorScheme.onPrimaryContainer
            )
        }
    }
  }
}

item {
  Column( horizontalAlignment =
    Alignment.CenterHorizontally
  ) {
    Text( text =
      userData.name,
      style = MaterialTheme.typography.headlineMedium, fontWeight
      = FontWeight.Bold
    )
    Spacer(modifier = Modifier.height(8.dp))
    Text( text = userData.email, style =
      MaterialTheme.typography.bodyLarge,
      color = MaterialTheme.colorScheme.onSurfaceVariant
    )
    Spacer(modifier = Modifier.height(8.dp))
    Text( text = if (userData.isActive) "Active" else
      "Inactive", style =
      MaterialTheme.typography.bodyMedium, color = if
      (userData.isActive)
        MaterialTheme.colorScheme.primary else
        MaterialTheme.colorScheme.error,
      modifier = Modifier
        .padding(horizontal = 16.dp)
        .padding(vertical = 4.dp)
        .background( color = if
          (userData.isActive)
            MaterialTheme.colorScheme.primaryContainer else
            MaterialTheme.colorScheme.errorContainer,
          shape = MaterialTheme.shapes.small
        )
```

```
                .padding(horizontal = 12.dp, vertical = 4.dp)
            )
        }
    }

    item {
        Card(modifier =
            Modifier.fillMaxWidth()
        ) {
            Column( modifier =
                Modifier
                    .fillMaxWidth()
                    .padding(16.dp)
            ) {
                Text( text = "User
                    Information",
                    style = MaterialTheme.typography.titleMedium, fontWeight
                    = FontWeight.Bold
                )
                Spacer(modifier = Modifier.height(12.dp))

                InfoRow("ID", userData.id)
                InfoRow("Name", userData.name)
                InfoRow("Email", userData.email)
                InfoRow("Status", if (userData.isActive) "Active" else "Inactive")
            }
        }
    }
  }
}
is ApiResponse.Error -> {
  Column( modifier =
    Modifier.align(Alignment.Center),
    horizontalAlignment = Alignment.CenterHorizontally
  ) {
    Text( text = "Error: ${(user as
      ApiResponse.Error).message}", color =
      MaterialTheme.colorScheme.error
    )
    Spacer(modifier = Modifier.height(8.dp))
    Button(onClick = { viewModel.loadUser(userId) }) {
      Text("Retry")
    }
```

```
                }
            }
        }
    }
  }
}

@Composable
private fun InfoRow(label: String, value: String) {
  Column {
    Text(  text  =
      label,
      style = MaterialTheme.typography.bodySmall,
      color = MaterialTheme.colorScheme.onSurfaceVariant
    )
    Text(  text  =
      value,
      style = MaterialTheme.typography.bodyMedium,
      modifier = Modifier.padding(top = 2.dp)
    )
    Spacer(modifier = Modifier.height(8.dp)) }
}
```

**AppNavigationSimple.kt**

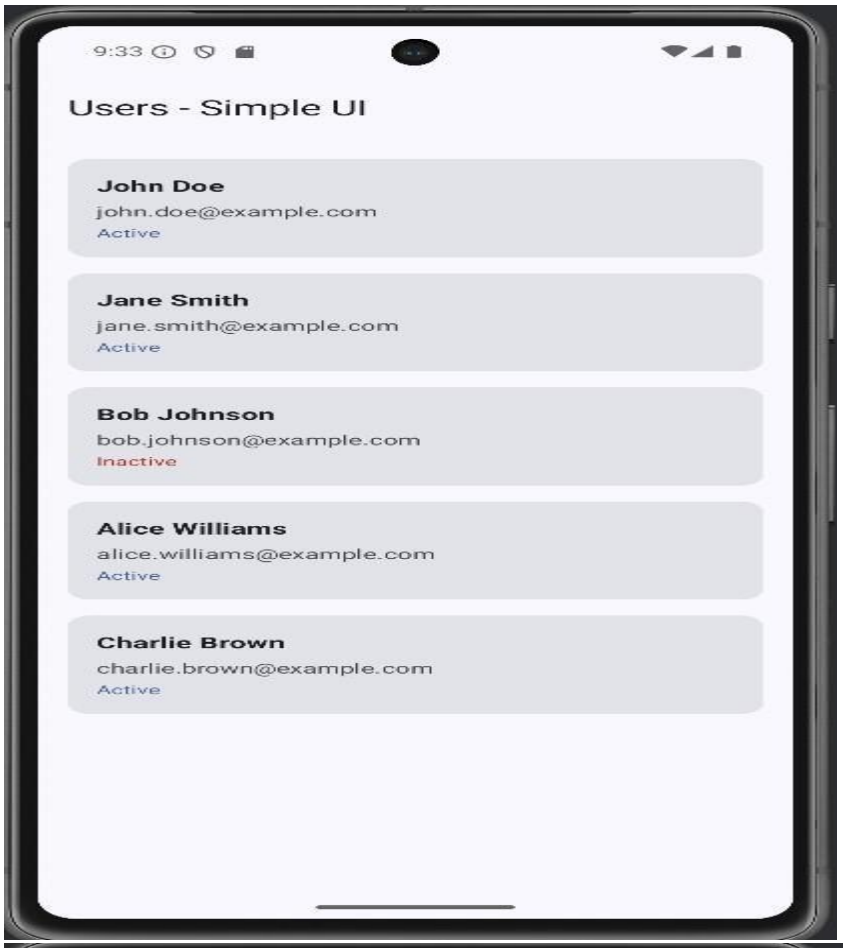
```
package com.example.apparchitectureuilayer.presentation.navigation
import androidx.compose.foundation.background import
androidx.compose.foundation.layout.*
import androidx.compose.foundation.shape.RoundedCornerShape
import androidx.compose.material3.* import
androidx.compose.runtime.* import
androidx.compose.ui.Alignment import
androidx.compose.ui.Modifier import
androidx.compose.ui.draw.clip import
androidx.compose.ui.text.font.FontWeight import
androidx.compose.ui.unit.dp
import com.example.apparchitectureuilayer.data.model.User
import com.example.apparchitectureuilayer.presentation.screen.UserListScreenSimple

@Composable
fun AppNavigationSimple() { var currentUser by remember {
  mutableStateOf<User?>(null) }

  if (currentUser == null) {
```

```
    UserListScreenSimple(
        onUserClick = { userId -> //
        Find user and show details val
        users = listOf(
                User( id = "1", name =
                    "John Doe",
                    email = "john.doe@example.com",
                    avatar = null,
                    isActive = true
                ),
                User( id = "2", name =
                    "Jane Smith",
                    email = "jane.smith@example.com",
                    avatar = null,
                    isActive = true
                ),
                User( id =
                    "3",
                    name = "Bob Johnson",
                    email = "bob.johnson@example.com",
                    avatar = null,
                    isActive = false
                ),
                User( id =
                    "4",
                    name = "Alice Williams",
                    email = "alice.williams@example.com", avatar
                    = null,
                    isActive = true
                ),
                User(
                    id = "5", name = "Charlie Brown", email
                    = "charlie.brown@example.com",
                    avatar = null,
                    isActive = true
                )
            )
            currentUser = users.find { it.id == userId } }
    )
} else {
    UserDetailScreenSimple(
        user = currentUser!!,
        onBack = { currentUser = null }
```

```
        )
    }
}

@OptIn(ExperimentalMaterial3Api::class)
@Composable fun
UserDetailScreenSimple(
    user: User,
    onBack: () -> Unit
) {
    Scaffold( topBar
        = {
            TopAppBar(
                title = { Text("User Details") }, navigationIcon
                = {
                    Button(onClick = onBack) {
                        Text("← Back")
                    }
                }
            )
        }
    ) { padding ->
        Column( modifier =
            Modifier
                .fillMaxSize()
                .padding(padding)
                .padding(16.dp), verticalArrangement =
            Arrangement.spacedBy(16.dp)
        ) {
            // User avatar placeholder
            Box( modifier =
                Modifier
                    .size(120.dp)
                    .align(Alignment.CenterHorizontally) .clip(RoundedCornerShape(16.dp))
                    .background( color =
                        MaterialTheme.colorScheme.primaryContainer
                    ),
                contentAlignment = Alignment.Center
            ) {
                Text( text = user.name.first().uppercase(), style =
                    MaterialTheme.typography.headlineLarge,
                    color = MaterialTheme.colorScheme.onPrimaryContainer
                )
```

```
        }

        // User info
        Card(                modifier                =
            Modifier.fillMaxWidth()
        ) {
            Column(  modifier  =
                Modifier
                    .fillMaxWidth()
                    .padding(16.dp),        verticalArrangement        =
                Arrangement.spacedBy(8.dp)
            ) {
                Text(        text        =        user.name,        style        =
                    MaterialTheme.typography.headlineMedium,
                    fontWeight = FontWeight.Bold
                )
                Text(        text        =
                    user.email,
                    style = MaterialTheme.typography.bodyLarge,
                    color = MaterialTheme.colorScheme.onSurfaceVariant
                )
                Text(  text  =  "Status:  ${if  (user.isActive)  "Active"  else
                    "Inactive"}",
                    style = MaterialTheme.typography.bodyMedium, color
                    = if (user.isActive)
                        MaterialTheme.colorScheme.primary else
                        MaterialTheme.colorScheme.error
                )
                Text( text = "ID: ${user.id}", style =
                    MaterialTheme.typography.bodySmall,
                    color = MaterialTheme.colorScheme.onSurfaceVariant
                )
            }
        }
    }
  }
}
```

**Output**


9:33
Users - Simple UI
John Doe
john.doe@example.com
Active
Jane Smith
jane.smith@example.com
Active
Bob Johnson
bob.johnson@example.com
Inactive
Alice Williams
alice.williams@example.com
Active
Charlie Brown
charlie.brown@example.com
Active

9:34
← Back
User Details
J
John Doe
john.doe@example.com
Status: Active
ID: 1

**Expt No : 9** **Date:**

## App Architecture (Persistence)

**Aim**

To store and retrieve data using Room persistence library.

**Algorithm**

1. Add Room dependencies.
2. Create Entity class.
3. Create DAO interface.
4. Create Database class.
5. Perform CRUD operations.

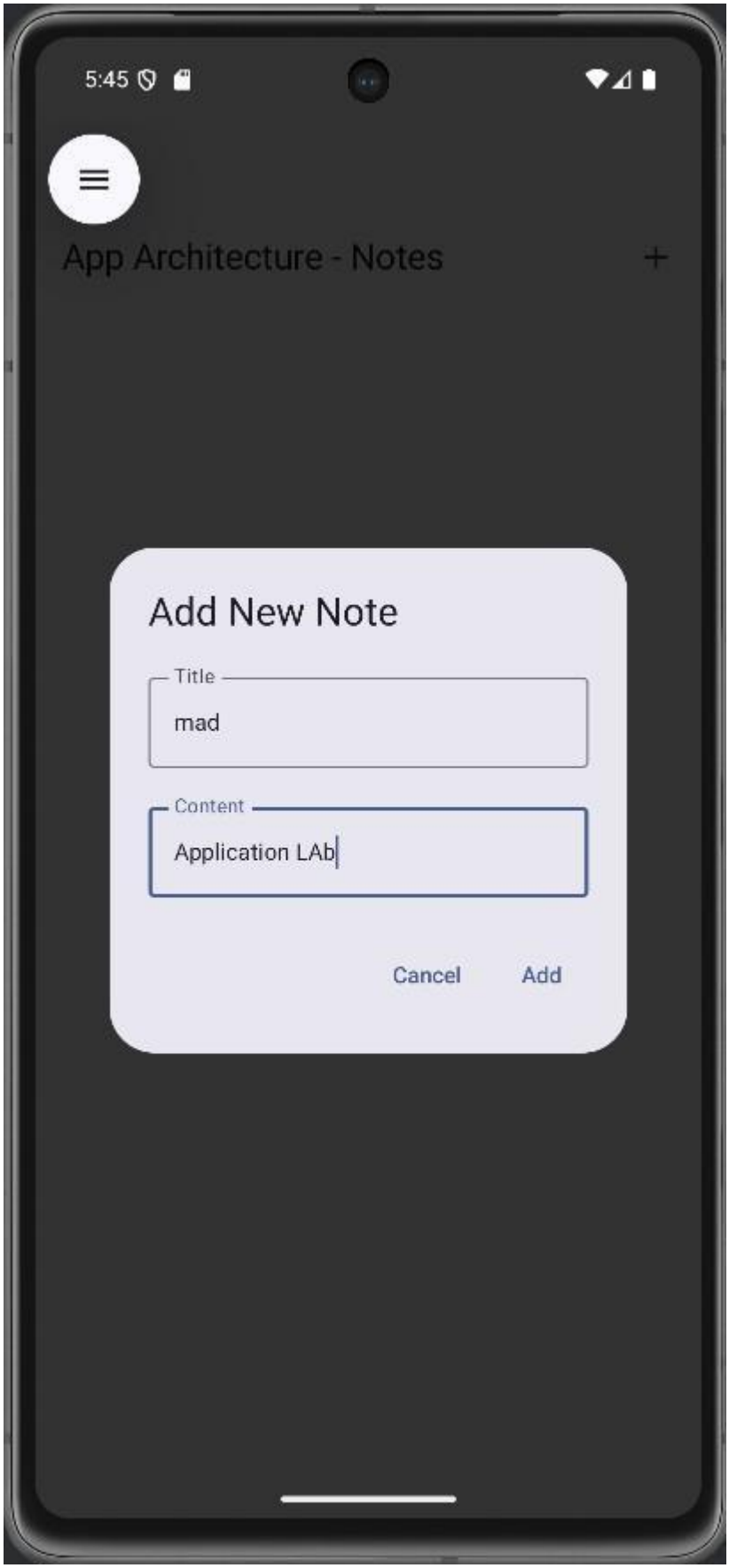
**MainActivity.kt**

```
package com.example.apparchitecturepersistence import
android.os.Bundle import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent import
androidx.activity.enableEdgeToEdge import
androidx.compose.foundation.layout.fillMaxSize import
androidx.compose.foundation.layout.padding import
androidx.compose.material3.Scaffold import
androidx.compose.runtime.Composable import
androidx.compose.runtime.collectAsState import
androidx.compose.runtime.getValue import
androidx.compose.ui.Modifier import
androidx.compose.ui.tooling.preview.Preview import
androidx.lifecycle.viewmodel.compose.viewModel import
androidx.compose.runtime.livedata.observeAsState import
com.example.apparchitecturepersistence.data.NoteDatabase
import com.example.apparchitecturepersistence.data.Note
```

```
import com.example.apparchitecturepersistence.repository.NoteRepository import
com.example.apparchitecturepersistence.ui.theme.AppArchitecturePersistenceThe
me import com.example.apparchitecturepersistence.ui.screens.NoteListScreen
import com.example.apparchitecturepersistence.viewmodel.NoteViewModel import
com.example.apparchitecturepersistence.viewmodel.NoteViewModelFactory

class MainActivity : ComponentActivity() { override
   fun onCreate(savedInstanceState: Bundle?) {
   super.onCreate(savedInstanceState)
   enableEdgeToEdge()

      val database = NoteDatabase.getDatabase(this) val
      repository = NoteRepository(database.noteDao())
      setContent {
         AppArchitecturePersistenceTheme {
            Scaffold(modifier = Modifier.fillMaxSize()) { innerPadding ->
               NoteApp(
                  repository = repository,
                  modifier = Modifier.padding(innerPadding)
               )
            }
         }
      }
   }
}

@Composable
fun NoteApp(repository: NoteRepository, modifier: Modifier = Modifier) {
   val viewModel: NoteViewModel = viewModel( factory =
   NoteViewModelFactory(repository)
   )
```

```
    val notes by viewModel.allNotes.observeAsState(initial = emptyList())
    val selectedNote by viewModel.selectedNote.observeAsState() val
    isLoading by viewModel.isLoading.observeAsState(initial = false) val
    errorMessage by viewModel.errorMessage.observeAsState()

    NoteListScreen( notes = notes ?:
        emptyList(), selectedNote =
        selectedNote, isLoading =
        isLoading ?: false,
        errorMessage = errorMessage,
        onNoteClick = { note -> viewModel.selectNote(note) }, onNoteDelete
        = { note -> viewModel.deleteNote(note) }, onNoteToggle = { note ->
        viewModel.toggleNoteCompletion(note) }, onAddNote = { title,
        content -> viewModel.insertNote(title, content) },
        onUpdateNote = { note -> viewModel.updateNote(note) },
        onClearError = { viewModel.clearError() }, onDeleteAll =
        { viewModel.deleteAllNotes() },
        formatTimestamp = { timestamp -> viewModel.formatTimestamp(timestamp) },
        modifier = modifier
    )
}

@Preview(showBackground = true)
@Composable fun
NoteAppPreview() {
    AppArchitecturePersistenceTheme {
        // Preview would need mock repository for full functionality
    }
}
```

Output




**Expt No : 10** **Date:**

## Advanced RecyclerView Use Cases

**Aim**

To display complex and dynamic lists using RecyclerView.

**Algorithm**

1. Add RecyclerView to layout.
2. Create ViewHolder.
3. Create Adapter.
4. Bind data list.
5. Handle item clicks.

```
package com.example.advancedrecyclerviewusecases

import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity
import androidx.recyclerview.widget.ItemTouchHelper
import androidx.recyclerview.widget.LinearLayoutManager
import androidx.recyclerview.widget.RecyclerView import
```

```
com.example.advancedrecyclerviewusecases.adapter.AdvancedRecyclerViewAdapter
import
com.example.advancedrecyclerviewusecases.helper.ItemTouchHelperCallbac
k import com.example.advancedrecyclerviewusecases.model.DataProvider
import com.example.advancedrecyclerviewusecases.model.Item import
com.google.android.material.snackbar.Snackbar class MainActivity :
AppCompatActivity() {

    private lateinit var recyclerView: RecyclerView private
    lateinit var adapter: AdvancedRecyclerViewAdapter private
    lateinit var itemTouchHelper: ItemTouchHelper private var
    items = DataProvider.getSampleData()

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        setupRecyclerView()
        setupSwipeToDelete()
        setupDragAndDrop()
    }
    private fun setupRecyclerView() {
        recyclerView = findViewById(R.id.recyclerView) adapter
        = AdvancedRecyclerViewAdapter(
            items = items, onItemClick
            = { item ->
                handleItemClick(item)
            },
            onDeleteItem = { item ->
                handleDeleteItem(item)
            }
        )

        recyclerView.apply {
            layoutManager = LinearLayoutManager(this@MainActivity)
            adapter = this@MainActivity.adapter
        }
    }

    private fun handleItemClick(item: Item) {
        Snackbar.make( recyclerView,
            "Clicked: ${item.title}",
            Snackbar.LENGTH_SHORT
```

```
    ).show()
}

private fun handleDeleteItem(item: Item) {
    val position = items.indexOf(item)
    if (position != -1) {
        items = items.toMutableList().apply { removeAt(position) }
        adapter.removeItem(item)

        Snackbar.make( recyclerView,
            "Deleted: ${item.title}",
            Snackbar.LENGTH_LONG
        ).setAction("Undo") { items = items.toMutableList().apply {
            add(position, item) }
            adapter.updateItems(items)
        }.show()
    }
}

private fun setupSwipeToDelete() {
    val callback = ItemTouchHelperCallback( onItemMove
        = { fromPosition, toPosition ->
            // Handle item move for drag and drop
            val movedItem = items[fromPosition] items
            = items.toMutableList().apply {
                removeAt(fromPosition)
                add(toPosition, movedItem)
            }
            adapter.notifyItemMoved(fromPosition, toPosition)
        },
        onItemSwiped  =  {  position  ->  //
            Handle item swipe for delete val
            swipedItem = items[position]
            handleDeleteItem(swipedItem)
        }
    )

    itemTouchHelper = ItemTouchHelper(callback)
    itemTouchHelper.attachToRecyclerView(recyclerView)
}

private fun setupDragAndDrop() {
    // Drag and drop is already set up in setupSwipeToDelete()
```

```
        // The ItemTouchHelper handles both swipe and drag operations
    }
}
```

**Output**

**Expt No :1 1** **Date:**

## Connect to the Internet

**Aim**

To fetch data from internet using REST API.

**Algorithm**

1. Add internet permission.
2. Add Retrofit dependency.
3. Create API interface.
4. Make network call.
5. Display response data.

**Coding**

**AndroidManifest.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools">
```

```
<application
android:allowBackup="true"
android:dataExtractionRules="@xml/data_extraction_rules"
android:fullBackupContent="@xml/backup_rules"
android:icon="@mipmap/ic_launcher"
android:label="@string/app_name"
android:supportsRtl="true"
android:theme="@style/Theme.SendEmail"
tools:targetApi="31">
<activity
android:name=".MainActivity"
android:exported="true">
<intent-filter>
<action android:name="android.intent.action.MAIN" />
<category android:name="android.intent.category.LAUNCHER" />
</intent-filter>
</activity>
</application>
</manifest>
```

**activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent" android:orientation="vertical"
tools:context=".MainActivity">
<TextView
android:id="@+id/tvEmail"
android:layout_width="match_parent"
```

```
android:layout_height="wrap_content
" android:text="To" /> <EditText
android:id="@+id/etEmail"
android:layout_width="match_parent"
android:layout_height="wrap_content
" android:ems="10"
android:inputType="textPersonName" />
<TextView
android:id="@+id/tvSubject"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Subject"    />    <EditText
android:id="@+id/etSubject"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="textPersonName"
/>
```

```
<TextView
android:id="@+id/tvMessage"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Message" />
<EditText android:id="@+id/etMessage"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="textPersonName" />
<Button android:id="@+id/btSend"
android:layout_width="match_parent"
```

```
android:layout_height="wrap_content
" android:text="Send"
android:textAllCaps="false" />
</LinearLayout>
```

**MainActivity.kt** package org.rajalakshmi.sendemail

```
import android.content.Intent import
androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
```

```
import android.widget.Button import
android.widget.EditText import android.widget.TextView
class MainActivity : AppCompatActivity() { override fun
onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_main) val etEmail :
EditText = findViewById(R.id.etEmail) val etSubject :
EditText = findViewById(R.id.etSubject) val etMessage :
EditText = findViewById(R.id.etMessage) val btSend :
Button = findViewById(R.id.btSend)
btSend.setOnClickListener { val email =
etEmail.text.toString() val subject =
etSubject.text.toString() val message =
etMessage.text.toString() val intent =
Intent(Intent.ACTION_SEND)
intent.putExtra(Intent.EXTRA_EMAIL, arrayOf(email))
intent.putExtra(Intent.EXTRA_SUBJECT, subject)
intent.putExtra(Intent.EXTRA_TEXT, message)
intent.type = "message/rfc822"
startActivity(Intent.createChooser(intent, "Choose an Email client
:"))
}
}
}
```

**Output**

**Expt No : 12** **Date:**

## Repository Pattern and WorkManager

**Aim**

To manage data sources and background tasks efficiently.

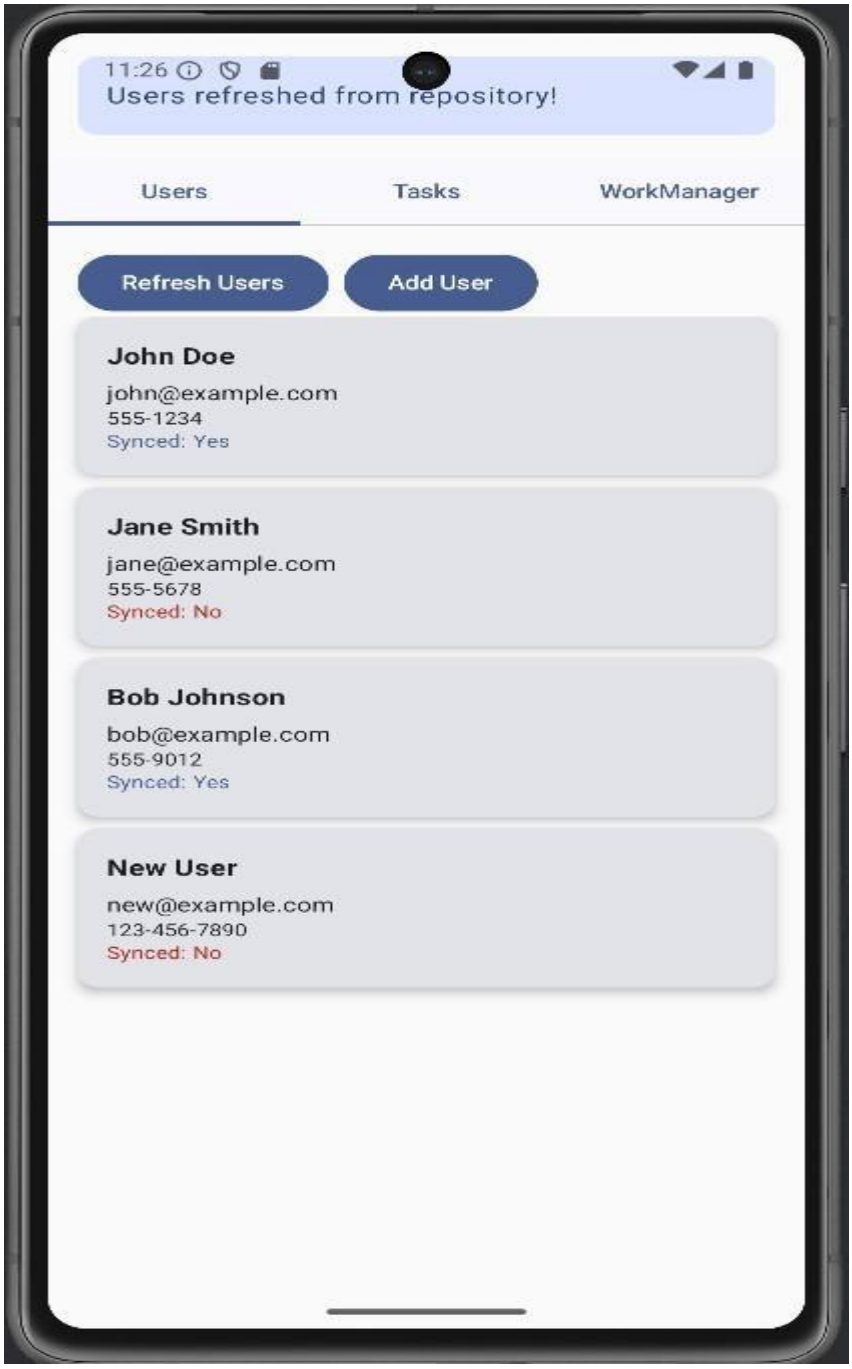
**Algorithm**

1. Create repository class.
2. Connect repository to ViewModel.
3. Add WorkManager dependency.
4. Create Worker class.
5. Schedule background task.

**Coding**

```
package com.example.repository

import android.os.Bundle
import androidx.activity.ComponentActivity import
androidx.activity.compose.setContent import
androidx.activity.enableEdgeToEdge import
com.example.repository.ui.theme.RepositoryTheme

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContent {
            RepositoryTheme {
                RepositoryDemoScreen()
            }
        }
    }
}
```

**Output**




**Expt No : 13** **Date:**

## App UI Design

**Aim**

To design attractive, responsive, and user-friendly Android UI.

**Algorithm**

1. Apply Material Design principles.
2. Use themes and styles.
3. Add colors and fonts.

4. Use icons and animations.
5. Test UI responsiveness.

MainActivity.kt

```
package com.example.japp9

import android.os.Bundle import
android.os.Handler import
android.os.Looper import
android.text.Editable import
android.text.TextWatcher import
android.widget.Button import
android.widget.EditText
import androidx.appcompat.app.AppCompatActivity import
androidx.recyclerview.widget.LinearLayoutManager import
androidx.recyclerview.widget.RecyclerView import
com.example.japp9.adapter.AdvancedRecyclerViewAdapter import
com.example.japp9.model.ExpandableItemModel import
com.example.japp9.model.RecyclerViewItem import
com.example.japp9.model.User import
com.example.japp9.utils.DragAndDropCallback import
com.example.japp9.utils.FilterUtils import
com.example.japp9.utils.InfiniteScrollListener import
com.example.japp9.utils.SwipeToDeleteCallback class
MainActivity : AppCompatActivity() {

    private lateinit var recyclerView: RecyclerView private
    lateinit var adapter: AdvancedRecyclerViewAdapter
    private lateinit var searchEditText: EditText private
    lateinit var toggleSwipeButton: Button private
    lateinit var toggleDragButton: Button

    private var allItems = mutableListOf<RecyclerViewItem>() private
    var currentItems = mutableListOf<RecyclerViewItem>()
    private var swipeToDeleteHelper: androidx.recyclerview.widget.ItemTouchHelper? =
null private var dragAndDropHelper: androidx.recyclerview.widget.ItemTouchHelper?
= null private var infiniteScrollListener: InfiniteScrollListener? = null

    private var isLoading = false private
    var currentPage = 1
    private val itemsPerPage = 10
```

```
override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView(R.layout.activity_main)

    initViews()
    setupRecyclerView()
    generateInitialData()
    setupListeners()
    setupGestures()
}

private fun initViews() { recyclerView =
    findViewById(R.id.recyclerView) searchEditText =
    findViewById(R.id.searchEditText) toggleSwipeButton =
    findViewById(R.id.toggleSwipeButton) toggleDragButton =
    findViewById(R.id.toggleDragButton)
}

private fun setupRecyclerView() { adapter =
    AdvancedRecyclerViewAdapter( items =
    currentItems, onItemClick = { item ->
    handleItemClick(item)
        },
        onDeleteItem = { position -> adapter.removeItem(position)
        }
    )

    recyclerView.layoutManager = LinearLayoutManager(this) recyclerView.adapter
    = adapter

    // Setup infinite scrolling
    val layoutManager = recyclerView.layoutManager as LinearLayoutManager
    infiniteScrollListener = object : InfiniteScrollListener(layoutManager) { override
    fun onLoadMore() { loadMoreItems()
        } }
    recyclerView.addOnScrollListener(infiniteScrollListener!!)
}

private fun generateInitialData() { //
    Generate sample users
    val users = (1..20).map { index ->
        User( id = index, name =
            "User $index",
            email = "user$index@example.com",
```

```
            avatar = "",
            isActive = index % 2 == 0
        )
    }

    // Generate sample expandable items val
    expandableItems = (1..5).map { index ->
        ExpandableItemModel( id =
            index + 100, title =
            "Category $index",
            description = "Description for category $index",
            subItems = (1..3).map { subIndex -> "Sub item $subIndex in category $index"
            }
        ) }

    allItems.clear() allItems.add(RecyclerViewItem.HeaderItem("Active
    Users"))
    users.filter { it.isActive }.take(5).forEach {
        allItems.add(RecyclerViewItem.UserItem(it))
    }

    allItems.add(RecyclerViewItem.HeaderItem("Categories"))
    expandableItems.forEach {
    allItems.add(RecyclerViewItem.ExpandableItem(it))
    }

    allItems.add(RecyclerViewItem.HeaderItem("All Users"))
    users.forEach {
    allItems.add(RecyclerViewItem.UserItem(it))
    }

    currentItems.clear()
    currentItems.addAll(allItems.take(itemsPerPage))
    adapter.updateItems(currentItems)
  }

  private fun setupListeners() { searchEditText.addTextChangedListener(object :
    TextWatcher { override fun beforeTextChanged(s: CharSequence?, start: Int, count:
    Int, after: Int)
{} override fun onTextChanged(s: CharSequence?, start: Int, before: Int, count: Int)
{} override fun afterTextChanged(s: Editable?) {
        filterItems(s?.toString() ?: "") }
    })
```

```
        toggleSwipeButton.setOnClickListener                    {
            toggleSwipeToDelete()
        }

        toggleDragButton.setOnClickListener                    {
            toggleDragAndDrop()
        } }

    private fun setupGestures() { // Setup swipe to delete val
        swipeCallback = SwipeToDeleteCallback { position ->
        adapter.removeItem(position)
        }
        swipeToDeleteHelper =
androidx.recyclerview.widget.ItemTouchHelper(swipeCallback)
        swipeToDeleteHelper?.attachToRecyclerView(recyclerView)

        // Setup drag and drop
        val dragCallback = DragAndDropCallback { fromPosition, toPosition ->
            adapter.moveItem(fromPosition, toPosition)
        }
        dragAndDropHelper =
androidx.recyclerview.widget.ItemTouchHelper(dragCallback)
        // Don't attach by default, let user toggle it
    }

    private fun handleItemClick(item: RecyclerViewItem) {
        when (item) { is RecyclerViewItem.ExpandableItem
        -> {
                // Toggle expandable item
                val updatedItem = item.item.copy(isExpanded = !item.item.isExpanded)
                val index = currentItems.indexOf(item) if (index != -1) {
                    currentItems[index] =
                RecyclerViewItem.ExpandableItem(updatedItem)
                adapter.notifyItemChanged(index) }
            }
            is RecyclerViewItem.UserItem -> {
                // Handle user click - could show details or edit println("Clicked
                on user: ${item.user.name}")
            }  else  ->
            {}
        }
    }
```

```
private fun filterItems(query: String) { val filtered =
FilterUtils.filterItems(allItems, query)
currentItems.clear()
currentItems.addAll(filtered.take(itemsPerPage))
adapter.updateItems(currentItems) }

private fun toggleSwipeToDelete() { if (swipeToDeleteHelper
  == null) { val swipeCallback = SwipeToDeleteCallback {
  position -> adapter.removeItem(position)
    }
    swipeToDeleteHelper =
androidx.recyclerview.widget.ItemTouchHelper(swipeCallback)
    swipeToDeleteHelper?.attachToRecyclerView(recyclerView)
    toggleSwipeButton.text = "Disable Swipe"
  } else {
    swipeToDeleteHelper?.attachToRecyclerView(null)
    swipeToDeleteHelper = null
    toggleSwipeButton.text = "Enable Swipe" }
}

private fun toggleDragAndDrop() { if (dragAndDropHelper == null) { val
  dragCallback = DragAndDropCallback { fromPosition, toPosition ->
  adapter.moveItem(fromPosition, toPosition)
    }
    dragAndDropHelper =
androidx.recyclerview.widget.ItemTouchHelper(dragCallback)
    dragAndDropHelper?.attachToRecyclerView(recyclerView)
    toggleDragButton.text = "Disable Drag"
```

```
    } else {
        dragAndDropHelper?.attachToRecyclerView(null
        ) dragAndDropHelper = null
        toggleDragButton.text = "Enable Drag"
    }
}

private fun loadMoreItems() { if
    (isLoading) return

    isLoading = true

    // Add loading indicator
    val loadingList = currentItems.toMutableList()
    loadingList.add(RecyclerViewItem.LoadingItem())
    adapter.updateItems(loadingList)

    // Simulate network delay
```

```
        Handler(Looper.getMainLooper()).postDelayed({
            val startIndex = currentPage * itemsPerPage
            val endIndex = (startIndex + itemsPerPage).coerceAtMost(allItems.size)

            if (startIndex < allItems.size) {
                currentItems.addAll(allItems.subList(startIndex, endIndex))
                currentPage++

                // Remove loading indicator and add new items
                adapter.updateItems(currentItems)
                infiniteScrollListener?.setLoaded()
            }

            isLoading = false
        }, 1500)
    }
}
```

10:07
App UI Designer
Components
Button
Text Field
Text
Image
Card
Container
Elements
BUTTON (element_1)
Properties
Text
Button
Width
100
Height
50
Corner Radius
8
Red
Blue
Green